Digging for Trouble?
Mining and Criminal Behavior of Young Males

Gabriel Rodríguez-Puello[1*]

[1]Jönköping International Business School (JIBS), Jönköping University

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Abstract

Does a higher wage deter crime or help finance it? Improved labor market conditions increase the opportunity cost of crime, but they also improve the potential payoff from committing crimes. This paper studies how a positive local labor market shock affects criminal behavior. I exploit the 2004 iron ore boom in Northern Sweden as an exogenous shock to local economic conditions, combining geocoded administrative data on all criminal convictions and demographics for 2000–2015 with a difference-in-differences design. Comparing young male residents of mining municipalities to those of similar nearby municipalities, I identify the causal impact of improved labor market opportunities on crime and characterize treatment effect heterogeneity using causal forests. The boom led to a 52 percent decline in property crime convictions among young male residents aged 18–29 (from a baseline rate of 1.2 percent), concentrated within 20 kilometers of the mines and driven primarily by first-time offenders. In contrast, substance-related crime convictions increased by 181 percent (from a baseline of 0.2 percent), particularly among repeat offenders and individuals directly employed in the mining sector. Violent and traffic crimes are unaffected. Mechanism evidence shows the boom raised local employment and earnings, while migration, policing intensity, and income inequality do not account for the results. (JEL R11, K42, Q33, O13)

Keywords: criminal behavior, economic opportunities, mining, Sweden

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1 Introduction

Does a higher wage deter crime or help finance it? The canonical economic model of crime (Becker, 1968; Ehrlich, 1973) offers a clear prediction: worse labor market conditions lower the opportunity cost of illegal activity, pulling individuals toward crime, while better conditions push them back toward legal work. A large empirical literature confirms the first half of this prediction: job loss and declines in wages increase criminal behavior (e.g., Britto et al., 2022). Evidence on the opposite direction, whether positive shocks reduce crime, is more limited and more mixed. Higher employment and earnings can substitute for crime by raising the returns to legal activity (e.g., Gould et al., 2002; Machin and Meghir, 2004), but the same framework also allows positive shocks to complement criminal activity by raising the returns to certain crimes or fueling demand for illegal goods (Draca and Machin, 2015; Cunningham et al., 2020). Whether positive economic shocks substitute for or complement crime, therefore, remains largely unresolved.

These opposing forces are unlikely to affect all crimes equally. Property crimes, closely tied to economic incentives, are most likely to respond to changes in the opportunity cost of illegal activity and should decline as legal opportunities improve (Becker, 1968; Ehrlich, 1973). Substance-related offenses may rise instead: higher earnings relax the budget constraint for substance purchases, raising demand for drugs and alcohol as consumption goods with positive income elasticity (James and Smith, 2017; Cunningham et al., 2020). Violent crimes, driven by interpersonal conflict and social norms rather than material incentives, are less likely to respond to economic shocks (Draca and Machin, 2015). Because different crimes are governed by different incentives, the net effect of a positive economic shock on aggregate criminal behavior is ambiguous, and aggregate crime measures can mask important heterogeneity.

This paper studies how a positive local economic shock affects criminal behavior, using the 2004 iron-ore boom in northern Sweden as a quasi-experimental setting. The boom was driven by global commodity prices, and therefore plausibly exogenous to local conditions, and concentrated in two municipalities, Gällivare and Kiruna, where mining accounts for about 20 percent of employment. I combine geocoded administrative registers covering all criminal convictions with rich individual-level information on employment, earnings, and residential location over the period 2000–2015. In addition to the administrative registers, I assemble an original text dataset from the Retriever Mediearkiv, consisting of newspaper articles mentioning the mining company LKAB. Administrative records capture what individuals do, but not the broader social conditions and working environment surrounding the boom. The newspaper corpus complements this data by tracking how media coverage of LKAB evolved month by month, allowing me to test whether the boom changed the discourse around working conditions and social outcomes in ways that complement and help interpret the patterns in the conviction data. The empirical design exploits spatial variation in exposure to mining activity and temporal variation induced by the global commodity price shock in a difference-in-differences framework, allowing me to isolate how improved local economic conditions shape criminal behavior while explicitly accounting for residential mobility and spatial heterogeneity. Moreover, the size and richness of the data set allow me to characterize the heterogeneity of treatment effects across individuals using causal forests.

Northern Sweden and the mining boom are ideal contexts for this study for several reasons. First, the commodity price shock was generated by global demand, particularly China's rapid industrialization, and speculative dynamics in international markets, making it plausibly exogenous to local crime trends and labor market conditions in Sweden (Radetzki et al., 2008; Farooki and Kaplinsky, 2013; Singleton, 2014). Second, the location of iron ore deposits is determined by geological conditions rather than local economic or social characteristics (Brunnschweiler and Bulte, 2009). Third, the shock was large enough to generate meaningful first-stage labor market effects. Previous empirical evidence shows that mining booms improve the labor market, attract in-migrants to the areas, and increase local purchasing power (e.g., Black et al., 2005; Rodríguez-Puello, 2025). In northern Sweden, prior work documents substantial increases in employment and earnings among residents of the treated municipalities following the boom, driven by the mining sector as well as by spillovers into manufacturing, construction, and services (Rodríguez-Puello and Rickardsson, 2026).

The mining boom produced opposing effects on different types of crime among young male residents aged 18–29. Property crime (e.g., theft, burglary, and fraud) convictions decline by 0.66 percentage points, a 52 percent drop relative to the pre-boom mean, while substance-related (e.g., narcotics possession and use) convictions increase by 0.46 percentage points, a 181 percent increase from a low baseline. Violent and traffic crimes are unaffected. Both effects are concentrated within 20 kilometers of the mines, with no significant changes for older cohorts aged 30–39. The magnitude of these effects is at the upper end of, but broadly consistent with, quasi-experimental estimates in the resource shocks literature (Online Appendix Figure B.1). The larger magnitudes partly reflect the focus on young males aged 18–29, who have the highest baseline crime rates and are the most responsive to changes in labor market conditions. Both effects are driven by pre-boom residents, not by compositional change through migration, though migrants to mining municipalities also show higher substance crime relative to those settling elsewhere.

Causal forest estimates show that these opposing responses operate on different individuals. On the one hand, the reduction in property crime is driven by first-time offenders and those with weak baseline labor market attachment, which are precisely the groups for whom improved economic opportunities represent the largest change in the returns to legal activity. On the other hand, the increase in substance-related crime is concentrated among repeat offenders, individuals employed in the mining sector, and those in the upper tail of the earnings distribution, pointing to an income and consumption channel: higher disposable income increases engagement in risky behaviors among individuals already involved in substance use, rather than inducing new entry.

To assess the welfare implications of these opposing crime responses, I translate the estimated effects into implied social costs by combining my estimates with existing estimates of the costs of crime (Heeks et al., 2018; Alsan et al., 2025). Reductions in property crime generate sizable local welfare gains, amounting to SEK 17.8 million during the 2004–2015 boom period (12 years), and reflecting lower victimization costs and reduced criminal justice expenditures. These gains are partially offset by social losses of roughly SEK 14.1 million from increased substance-related crime, which impose social costs through enforcement, health, and broader social harms.

Mechanism analysis reveals important insights. The mining boom substantially increased employment and income among young male residents, raising the opportunity cost of property crime (legal wages now exceed criminal returns) and shifting the cost-benefit calculation away from criminal activity, especially for low labor market attachment males (Becker, 1968). By contrast, for mining-employed males, higher income relaxes the budget constraint for engagement in risky behaviors, specifically drug consumption. This pattern is not a mining-specific finding, as it is consistent with a well-documented tendency for workers in dangerous, physically demanding occupations to exhibit elevated substance use when income rises (Frone, 2006; Bush and Lipari, 2015). An original corpus of LKAB newspaper articles, analyzed using an interrupted time series, finds no shift in stress-related coverage at the boom's onset, ruling out occupational stress. Migration, income inequality, and police capacity do not account for the results. I discuss each ruled-out alternative in Section 6.1. The estimates are also robust to several changes in assumptions and estimation.

Related literature. This paper contributes to several strands of literature on crime, labor markets, and local economic shocks. First, this paper contributes to the literature on crime and local economic conditions. I provide micro-level causal evidence that the same positive shock can reduce economically motivated crime while increasing consumption-driven offenses, helping reconcile the mixed findings in the resource-shocks literature (e.g., James and Smith, 2017; Axbard et al., 2021; Street, 2025).[1] The substance-crime increase is consistent with an income channel established in the broader labor-and-health literature on workers in physically demanding, well-paid occupations (Frone, 2006; Bush and Lipari, 2015). The implied earnings elasticity for property crime (approximately -2) is at the upper end of existing estimates

[1]Studies exploiting resource-driven booms often find increases in aggregate crime, especially in the context of oil and gas extraction and fracking in the United States (e.g., Raphael and Winter-Ebmer, 2001; Gould et al., 2002; James and Smith, 2017; Dix-Carneiro et al., 2018). More recent micro-level work, however, shows that improved labor market opportunities generated by resource shocks can reduce criminal behavior, consistent with rational models of crime (e.g., Axbard et al., 2021; Street, 2025). There is also a relatively small literature specifically on mining booms and crime, where existing studies yield mixed results and most rely on aggregate outcomes (e.g., Carrington et al., 2011; Corvalan and Pazzona, 2019).

(Gould et al., 2002; Machin and Meghir, 2004), suggesting the opportunity cost channel is particularly strong in this setting. Evidence from developed-country settings is particularly scarce (Komarek, 2018), where the institutional environment allows for a cleaner test of the economic channel. The Swedish setting — a stable democracy with strong institutions — isolates the economic channel from the armed-conflict and weak-state confounders that characterize much of the developing-country evidence (Collier and Hoeffler, 2005; Berman et al., 2017).

Second, I also contribute to the literature on residential mobility and crime. By distinguishing residents from migrants in individual-level panel data, I show that aggregate crime patterns following resource shocks reflect compositional change as much as behavioral response, complementing Street (2025) for US fracking. Focusing on individuals rather than places allows me to separate behavioral responses from selection effects driven by population mobility.[2] In addition, the richness of the panel further allows me to characterize effects across crime types, economic sectors, time, and demographic groups, providing more granular evidence than aggregate crime measures permit.

Finally, the paper provides new evidence on heterogeneity in crime responses to economic opportunities and contributes to the growing literature using machine learning methods in empirical economics. I use causal forests estimators (Athey and Imbens, 2016; Wager and Athey, 2018; Athey et al., 2019) to let the data identify the relevant dimensions of treatment-effect variation and characterize treatment effect heterogeneity across individuals, an approach still rare in the crime-and-economic-shocks literature. Together with an original LKAB newspaper corpus from the Retriever Mediearkivet press archive and an interrupted time series test of the occupational stress channel, these tools allow a richer characterization of who responds to the boom and through which mechanism than is possible from average effects alone.

Roadmap. The remainder of the paper is structured as follows. In Section 2, the background of the Swedish mining sector and the mining booms are presented. Section 3 presents the data and sample. Section 4 presents the empirical framework and identification assumptions. Section 5 reports the main results, the heterogeneity analysis using causal forests, and the migrant analysis. Section 6.1 translates the estimates into welfare effects and discusses the mechanisms behind the results. Section 7 concludes.

2 Background and institutional setting

This section provides background on the mining boom, the mining sector in Sweden, and its local economic effects, including a summary of previous evidence on the labor market effects

[2] A growing literature emphasizes the importance of separating individual behavioral responses from compositional changes when evaluating local economic shocks (Guettabi and James, 2020; Kovalenko, 2023; Jacobsen et al., 2023).

of mining booms in Sweden.

In the last two decades, resource-dependent countries and mining communities have experienced the economic and socioeconomic impacts of resource shocks in the form of price booms. These are characterized by large and persistent increases in international prices of minerals (Fleming and Measham, 2015; Álvarez et al., 2021). I analyze the global mining boom that started during the first years after the new millennium (2004) when international mining prices suddenly tripled (Baffes and Haniotis, 2010). It is difficult to choose the timing of the mining boom because of the complex fluctuations of international prices of different minerals (Rossen, 2015). Following Tano et al. (2016), I use 2004 as the starting point because it is the year when the price of minerals started to rapidly increase; for example, the price of iron ore increased by 67% from 2004 to 2005. In addition, the number of mining jobs had a negative trend until 2003, started to increase in 2004, and continued to grow over the coming years (SGU, 2014; Knobblock and Pettersson, 2010). This trend was accompanied by an increase in investment in the Swedish mining sector. According to the literature, this shock can be considered a quasi-experiment and plausibly exogenous if it fulfills four conditions: large, variable, temporary, and generated outside an industry or country. It was generated by China's increasing demand for commodities (Radetzki et al., 2008; Farooki and Kaplinsky, 2013) and speculation in stock markets that generated investor flow (Singleton, 2014; Erten and Ocampo, 2013), rather than shifts in the supply of minerals. Therefore, it was generated outside of the country. In addition, it must be large and variable enough to affect municipalities' local conditions and temporary to identify the phases and years in which it occurred. Since this external demand shock is exogenous to the Swedish mining industry, it allows me to identify causal effects of labor market shocks on criminal behavior. Moreover, being able to track individuals over a long period provides a unique setting to examine how criminal behavior responds to changes in local economic conditions.

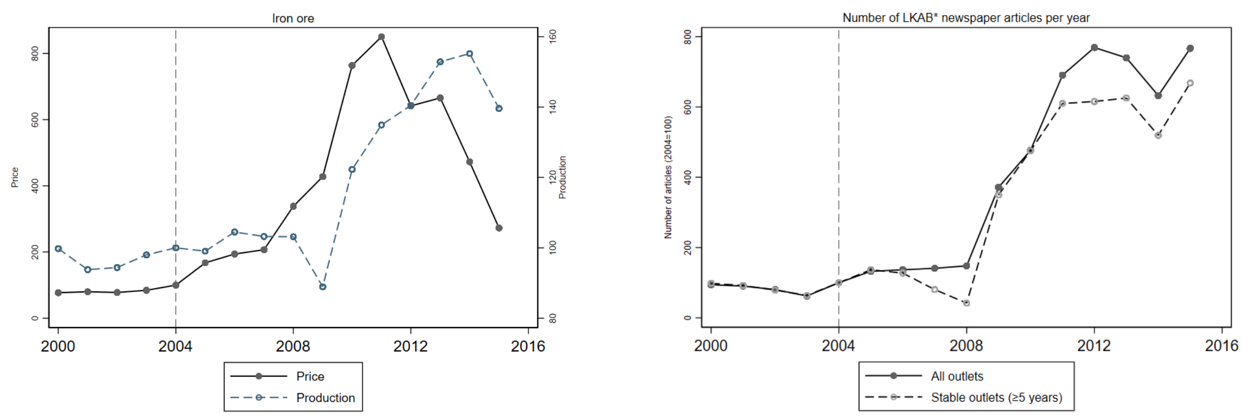
This boom is especially relevant for Sweden because the country has a long tradition of mining. During the mining boom, the main minerals and metals exploited in the country were iron ore, copper, zinc, and gold (Tano et al., 2016). I focus on iron ore because it is the most important mineral in the Swedish mining economy, in which the country is dominant at the European level, producing approximately 90% of the total iron ore production in the European Union (SGU, 2016). Figure 1, panel (a), shows the international prices and Swedish production of iron ore for the period 2000-2015. As can be seen, prices began to increase in 2004, reaching the maximum level in 2011. The price of iron ore increased by 67% from 2004 to 2005 and continued to grow rapidly in the following years (Tano et al., 2016). At the same time, observing the rise in prices, mining companies employed strategies to increase their production before a probable fall in prices, showing some changes in production between those same years after the increase in prices. In addition, the start of the mining boom coincides with a dramatic increase in exploration activities and production in Northern Sweden due to high local and international investment in the sector and increasing demand for minerals and metals (Petterson and Knobblock, 2010; SGU, 2014).

In addition to changes in prices and production, the mining boom was highly salient in the public debate. Using information from the newspaper archive *Retriever Mediearkivet*, I construct the annual number of articles in Swedish newspapers mentioning LKAB (Luossavaara–Kiirunavaara Aktiebolag), the state-owned iron ore company operating the major mines in Kiruna and Gällivare (Figure 1, panel (b)). Media coverage closely tracks the evolution of international iron ore prices, increasing sharply after 2004 and peaking during the height of the boom.[3] This pattern suggests that the mining boom was widely perceived and discussed at the national level, reinforcing the view that the shock was not only economically significant but also highly visible to local communities.

Figure 1: The 2004 iron ore mining boom in Sweden, 2000–2015

(a) Price and production of iron ore

(b) Media coverage of LKAB in newspapers

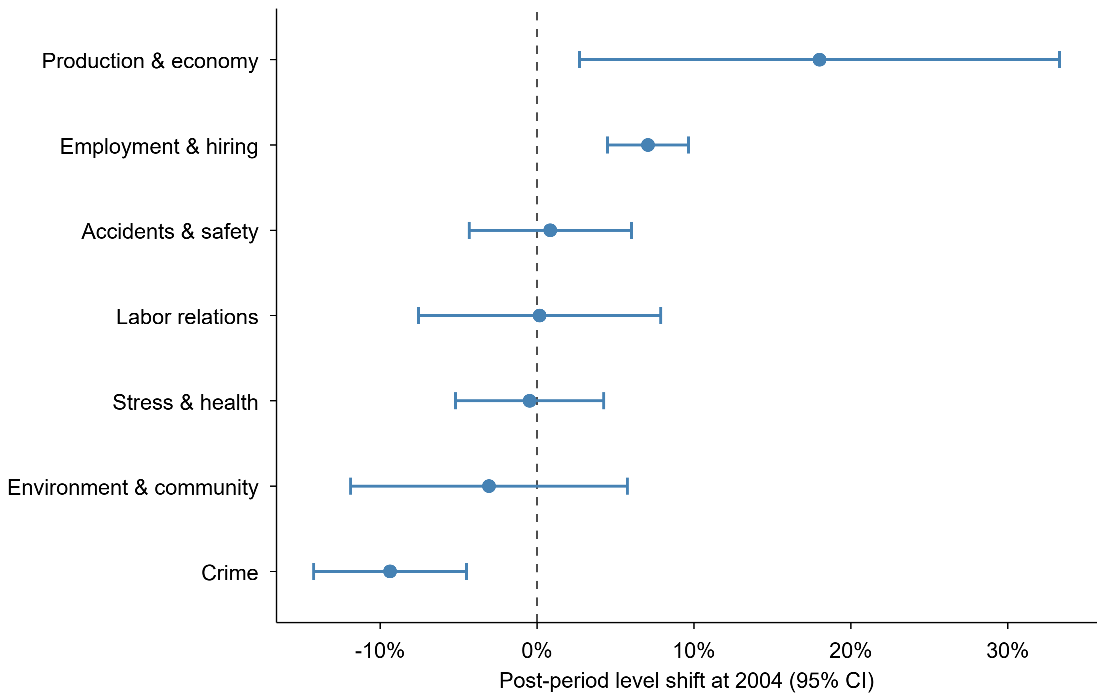


Notes: Both series are normalized to 2004 values (2004=100). The vertical dashed line marks the start of the mining boom (2004). *Panel (a):* The figure plots the international iron ore price (solid line, left axis) and Swedish iron ore production volumes (dashed line, right axis) over 2000–2015. The price series is from the International Monetary Fund; the production series is from the Geological Survey of Sweden (SGU, 2021). *Panel (b):* The solid line reports the annual number of newspaper articles mentioning "LKAB" normalized to 2004 values (2004=100). The dashed line restricts to outlets present in at least five distinct years of the panel. Articles are identified using the newspaper archive *Retriever Mediearkivet*. Following the literature, we search for articles containing the case-insensitive string "LKAB*", where the asterisk is used as a wildcard. The counts reflect exported and parsed articles. The vertical dashed line marks the start of the mining boom in 2004. LKAB is the main iron ore producer in Sweden and operates the large-scale mines in Kiruna and Gällivare. LKAB is Sweden's state-owned iron ore mining company.

[3] A potential concern is that the rise in LKAB article counts after 2004 partly reflects the expansion of Retriever's coverage, where more outlets were indexed each year, mechanically inflating article totals. The fact that the number of articles was on a declining trajectory in the years before the boom speaks against this interpretation. However, I acknowledge that the rapid surge after 2008 is driven by both an increase in the number of articles per outlet and an increase in the number of distinct outlets covering LKAB. Finally, the dashed line in the Figure shows that restricting the sample to outlets present in at least five years of the panel leaves the upward trend essentially unchanged.

The boom did not just change the economic landscape of northern Sweden; it changed what people read about LKAB. Using a a topic decomposition of the article corpus and classifying articles into seven thematic categories using keyword matching, Figure 2 plots, for each topic, the estimated level shift in the monthly share of LKAB articles at January 2004, that is, how much more (or less) of the coverage each topic captured after the boom onset relative to the pre-boom period.[4] As can be seen, the coverage shifted sharply toward production and economic output and employment and hiring at the boom onset, while crime fell and other topics remained flat, consistent with the boom narrative of expansion and hiring. Coverage of stress and mental health, accidents and safety, and labor relations shows no significant level shift at 2004, while crime coverage falls significantly (Online Appendix Table F.2).

Figure 2: Topic composition of LKAB newspaper coverage, 2000–2015



Notes: Each point is the estimated post-period level shift ($\hat{\beta}_1$) from an interrupted time series regression of the monthly share of LKAB*-related newspaper articles mentioning keywords from the indicated topic category, with a structural break at January 2004. Horizontal lines show 95% confidence intervals based on Newey–West standard errors (lag = 12). Topics are ordered by point estimate. The keyword dictionary is reported in Online Appendix Table F.1; full regression estimates in Online Appendix Table F.2; quarterly trends in Online Appendix Figure F.1. Articles are identified using the newspaper archive *Retriever Mediearkivet*. LKAB is Sweden's state-owned iron ore company operating large-scale mines in Kiruna and Gällivare.

In Sweden, mining activity is spatially concentrated in northern municipalities, with

[4] Each coefficient is estimated from a separate interrupted time series (ITS) regression of the monthly share of articles mentioning topic-specific Swedish keywords on a post-period indicator, a linear time trend, and a post-boom slope change, with Newey–West standard errors (lag = 12). Horizontal bars are 95% confidence intervals. The full description of the data and method is in Online Appendix F. The keyword dictionary is reported in Online Appendix Table F.1; full regression estimates in Online Appendix Table F.2.

a few exceptions in the South of the country. The North of the country is part of the *Fennoscandinavian Shield*, a region considered rich in minerals (Nordregio, 2009; Haley et al., 2011). Most mines, mining jobs, and exploration are concentrated in the two northernmost counties: Norrbotten and Västerbotten (SGU, 2014), representing 93% of total mining employment in Sweden in 2013 (Moritz et al., 2017). There are mainly three large iron ore mines that were continuously operating during the mining boom period: the Malmberget mine located in Gällivare municipality and the Kirunavaara and Gruvberget mines in Kiruna municipality.[5] These are all existing mines, with Kirunavaara opening in the 1860s, and Malmberget in the 1820s. I focus on existing mines instead of the opening or closing of mines since that was rare during this period and does not provide sufficient variation for empirical analysis. Moreover, these mines are central to the labor market dynamics of these municipalities, employing a substantial share of the workforce.

Due to the lack of an official classification for mining and non-mining municipalities in Sweden, I identify treated municipalities as those highly specialized in mining, with a high mining employment share in 2003, focused on the exploitation of iron ore: Gällivare and Kiruna. Both municipalities had at least one operating iron ore mine during the boom period, with the mining sector representing more than 10% of total employment. Choosing treated units based on their employment share in the industry is a common approach in the empirical literature on resource booms (Black et al., 2005; Kumar, 2017; Jacobsen et al., 2023). To ensure that treated and control individuals are not only similar but also geographically close, I define the control group as individuals located in other municipalities within Norrbotten County.[6]

Gällivare and Kiruna are small municipalities with approximately 18,000 and 23,000 residents in 2015, respectively, characterized by very low population density and compact urban centers surrounded by sparsely populated land. In both municipalities, large-scale iron ore mines are located in close proximity to the urban centers, and local commuting patterns, employment opportunities, and economic activity are tightly linked to mining operations. Although neither municipality is a classic company town, mining represents the dominant economic sector in both Gällivare and Kiruna, directly and indirectly employing a substantial share of the local workforce. Both municipalities had experienced decades of population

[5]There are other small mines in other municipalities, not considered in the study due to their size and because they are located in different parts of Sweden in terms of demographics and labor market. Other than Gällivare and Kiruna, the other eight municipalities that have mines during the mining boom period are Lycksele, Malå, Norsjö, Skellefteå, Sorsele, and Storuman in Västerbotten County, Askersund in Örebro County, and Hedemora in Dalarna County. Online Appendix Table A.1 shows some basic information about the mines, municipalities, and their employment share in the mining sector. Tano et al. (2016) and SGU (2021) provide a more detailed description of the mines opening and closing in Sweden, the public and private companies operating them, and the locations of the mines.

[6]Norrbotten County has 14 municipalities: Arjeplog, Arvidsjaur, Boden, Gällivare, Haparanda, Jokkmokk, Kalix, Kiruna, Luleå, Pajala, Piteå, Älvsbyn, Överkalix, and Övertorneå.

decline and disinvestment before the boom (Adjei et al., 2023), making the labor market shock particularly large relative to the local economic baseline. Moreover, prior to the mid-2000s mining boom, crime rates in both Gällivare and Kiruna were relatively stable and comparable to those in similar northern municipalities, with no pronounced differential trends relative to the control areas used in the analysis. Although the institutional environment of Swedish mining is specific, Gällivare and Kiruna share features with mining towns worldwide, such as copper-mining municipalities in northern Chile (Corvalan and Pazzona, 2019; Rodríguez-Puello, 2025), including geographic concentration of extraction activity, reliance on mainly one economic sector, and high sensitivity to global commodity price fluctuations, suggesting that comparative analyses across mining regions could help assess the external validity of the findings.

A growing body of evidence documents the labor market effects of mining booms in Sweden. Tano et al. (2016) finds positive effects on employment and labor income in Swedish mining municipalities among mining and construction workers, while Moritz et al. (2017) documents population growth and improved labor market conditions in these areas. Haikola and Anshelm (2020) highlights how the volatility of global iron ore prices influenced local attitudes toward state involvement and economic policy in mining communities. More recently, Rodríguez-Puello and Rickardsson (2026) use individual-level data to show that the mining boom significantly increased employment and earnings for residents in mining areas, with particularly strong effects in the mining sector. Moreover, the effects spread across space, sectors, and demographic characteristics. For example, the authors find that the mining boom affects the labor market conditions of individuals located as far as 27 km during the boom and 83 km in later years. Residents living near mines experienced around 5% higher annual earnings, equivalent to roughly 8,400 SEK in 2004. These findings provide the basis for examining whether improved labor market conditions constitute a channel through which the mining boom affects criminal behavior.

3 Data and sample

To examine the role of the mining boom on criminal behavior in Sweden, I rely on geocoded register data that originate from various administrative registers managed by Statistics Sweden. The data are of yearly frequency, and the outcomes are measured in November each year. The dataset is rich and contains information on all individuals above the age of 16, including age, gender, education, region of origin, income, and household characteristics. The data also include information on employment, occupation, economic sector, and region of residence and work, and I focus on the period 2000-2015. The analysis ends in 2015 for two reasons. First, extending the sample well beyond the boom period risks introducing noise, as the effects of the shock are expected to dissipate, and other local economic shocks may confound the estimates. Second, Europe and Sweden experienced a migration crisis from around 2015, with refugees disproportionately placed in peripheral and rural areas (Wennström and Oner, 2019; Puschmann et al., 2019); extending the sample would risk confounding the effects of the mining boom with those of this demographic shock. These data have been linked to the Swedish Conviction Register, maintained by the National Council for Crime Prevention (Brottsförebyggande rådet - BRÅ). These data contain comprehensive details concerning criminal convictions at the individual level during this period. It includes information on the type of crime and the date of the crime, among other information. A single conviction may encompass multiple crimes, and I observe all crimes within a given conviction. It excludes minor offenses such as speeding tickets, but includes offenses such as driving without a license and DUI.

I restrict the sample to young males: males older than 18 years and under 29 years who appear in five or more annual observations consecutively in the sample. Young males are an important population group to analyze the effect of economic opportunity on individuals' criminal behavior for several reasons. First, it is well established in previous empirical literature on crime that conviction rates are substantially higher among males than females, and they peak in early adulthood before declining steadily with age (e.g., Elonheimo et al., 2014; Epper et al., 2022). Importantly, young males are not only the group with the highest baseline crime rates, participating in a disproportionate amount of violent and property crimes (Komarek, 2018), but also the most responsive to changes in criminal incentives (Anker et al., 2021). Second, the mining sector worldwide and in Sweden is composed primarily of young and male workers (e.g., Kearney and Wilson, 2018; Chávez and Rodríguez-Puello, 2022). Finally, Rodríguez-Puello and Rickardsson (2026) finds evidence that the benefits from the mining boom in Sweden, through higher earnings and more employment opportunities, are large for males and young individuals located close to the mines.[7]

As described in Section 2, I define individuals located in Gällivare and Kiruna as treated and those in other Norrbotten municipalities as controls. Finally, I exclude individuals who moved to Norrbotten County in 2004 or later in the main specification, whom I call migrants. Therefore, the main analysis focuses on residents. I assume that those who migrated to this area after the shock did so in response to improved labor market conditions. This is important because the results may be a combination of the effects of the mining boom on crime and endogenous movement decisions made by individuals who migrated to the mining areas (Winters et al., 2021). Nevertheless, for robustness, I also present the results with all

[7] Online Appendix Figure B.2 shows the conviction rates by gender and age as a descriptive pattern for the Swedish data, and all types of crimes. In both the pre-boom and boom periods, young males stand out as the group with the highest conviction rates, several times greater than those of women or older cohorts. Therefore, young males are disproportionately responsible for overall crime levels, and most variation in criminal activity is concentrated in this demographic. I also examine males aged 30–39 as a falsification exercise, as shown in Online Appendix C. There are no statistically significant effects of the mining boom on criminal behavior for this older group across any crime type.

individuals. Online Appendix Figure B.3 shows the spatial location of the treated and control municipalities.

Individuals residing closer to the mines may be more affected by the mining boom than those further away, even within the treated municipalities. To examine this spatial heterogeneity, I construct a measure of the distance in kilometers from each individual's residential location to the nearest mine, based on the coordinates of the grid where he or she is located. I consider the three large iron ore mines that were continuously operating during the mining boom period, as mentioned in Section 2. The grids in the data are 250 by 250 meters in size in urban areas and 1000 by 1000 meters in size in rural areas. Individuals are located in these grids according to their place of residence. I classify individuals into 20-kilometer rings based on their proximity to the nearest mine, obtaining five rings in total, where those located farther away serve as controls. This approach, commonly known as the "ring method" (e.g., Wilson, 2012; Benshaul-Tolonen et al., 2019; Rodríguez-Puello and Rickardsson, 2026), exploits within-municipality variation in exposure and allows me to test whether effects decay with distance from the mines, a pattern expected if the mining boom is the true driver of the results.[8]

In the main analysis, the main outcomes of interest that reflect the criminal behavior of individuals are: (1) being convicted of property crime, (2) violent crime, (3) traffic crimes, and (4) substance-related crimes, per year. Property crimes include theft, robbery, and other assaults, fraud and other misconduct, embezzlement and other faithlessness, offenses against creditors, and crimes of damage. Violent crimes include violations of life and health, violations of freedom and peace, defamation, sexual offenses, and crimes against family. Traffic crimes include a broad range of road- and maritime-traffic offenses such as reckless or negligent driving, unlawful driving, driving under the influence, hit-and-run, speeding, safety violations, and other traffic- and navigation-related offenses.[9] Substance-related crimes include convictions under Swedish legislation governing narcotics, doping substances, alcohol, tobacco, and nicotine products, covering offenses related to possession, use, production, and distribution of these substances.[10] See Table A.2 in the Online Appendix for a detailed description of each category and subcategory of crime.[11] Online Appendix Table A.3 presents summary statistics for treated and controls before (2000-2003) and after

[8] Online Appendix Figure B.4 shows the distribution of individuals in space according to their location and distance to the nearest mine and their distribution in the rings.

[9] Traffic crime is identified using convictions under the Traffic and Maritime Offences Acts. The data do not allow for a consistent separation between driving under the influence and other traffic offenses within this category.

[10] The administrative conviction records do not allow for a consistent separation between possession, use, and distribution within narcotics offenses. As a result, substance-related crime is analyzed as a composite outcome.

[11] Other types of crimes, such as crimes against the public, crimes against the state, and other special categories (e.g., smuggling, tax crimes, terrorist crimes), are excluded due to their low frequency.

(2004-2015) the mining boom. The two groups are balanced in terms of demographics and job characteristics, confirming the expectations about the similarity of individuals in the treatment and control groups before the boom. The summary statistics show similar crime levels among treated and controls. The main sample consists of 27,525 individuals (440,402 individual-year observations). Moreover, I identify 15,108 migrants (241,725 migrant-year observations).

4 Empirical framework

Understanding the effects of economic opportunity from mining on criminal behavior is challenging from an empirical standpoint. Empirical literature often argues that resource endowments are exogenous because they occur due to chance and not so much to the political and economic environment in the host country. Therefore, according to Brunnschweiler and Bulte (2008) and Van der Ploeg and Poelhekke (2010), they are considered good measures of exogenous variation in resource wealth. Nevertheless, recent literature challenges this view. Although the location of natural resources is considered exogenous since it is determined by local geology and natural characteristics (Brunnschweiler and Poelhekke, 2021), the opening of mines and finding these resources is not completely exogenous, as it often relies on foreign firms to provide capital and expertise or on local governments' investment in exploration (Cust and Harding, 2020; Brunnschweiler and Poelhekke, 2021). Moreover, minerals are typically found in places that are remote and rugged; therefore, settlements driven by natural resources are highly heterogeneous from other communities (Asher and Novosad, 2023). Those features are not orthogonal to the labor market conditions or crime; hence, regressions of such outcomes on mining availability may be biased due to omitted variables and/or reverse causality.

I exploit the unexpected rise in iron ore prices that generated the global mining boom in 2004, coupled with variation in individuals' exposure to mining activity, driven by their geographical residential location. These provide a plausibly exogenous shock to local economic conditions, based on the assumption that the location of mines is exogenous because it depends on the local geology (Pelzl and Poelhekke, 2021; Christian and Barrett, 2024). Using a generalized difference-in-differences framework that exploits both temporal and spatial variation in exposure, I compare the criminal behavior of treated individuals to residents in other municipalities in the county, before and after the mining boom, to identify the average treatment effect (ATE) on the treated in mining municipalities. Formally, I estimate the effects of local economic shocks on local residents' criminal behavior using the following linear probability model:[12]

[12] While logit and probit models are also used for binary outcomes, they add their own assumptions, often don't have closed-form solutions, and their interpretation is more complex, especially with large amounts of fixed effects (Huntington-Klein, 2021).

$$Y_{ijmt} = \alpha_i + \alpha_j + \alpha_t + \beta(Post_t \times Treated_{imt}) + \epsilon_{ijmt} \quad (1)$$

where Y_{ijmt} is equal to 1 if individual i located in grid j and in municipality m in year t is convicted of a crime (property, violent, traffic, or substance crime). $Treated_{imt}$ is a binary variable that takes the value of 1 if individuals are located in the mining municipalities and 0 if individuals reside in other municipalities in the county (control). $Post_t$ is a binary indicator equal to 0 before the mining boom (2000-2003) and 1 after (2004-2015). The coefficient of interest is the β, which identifies the difference-in-differences estimate (ATE) of the effects of the mining boom on the outcome Y_{ijmt}. I include α_i, α_j, and α_t, which are individual, grid, and time fixed effects, respectively, to account for omitted variables and isolate the effect of the event. Individual fixed effects account for any static differences in the propensity to commit a crime across individuals. Year fixed effects control for factors that affect the criminal behavior of all individuals in a given year, such as the Great Recession. Grid fixed effects account for any static differences in the propensity to commit a crime across geographical locations.[13] In all estimations, I cluster standard errors at the grid level, allowing for an arbitrary covariance structure over time within each grid, and accounting for the serial correlation in the error term (Bertrand et al., 2004; Miller, 2023). In addition to the municipality-level treatment, I exploit fine-grained spatial variation in distance to the mines to explore spatial heterogeneity in the ATE and examine whether the effects decay with proximity, as expected if the mining boom is truly driving the results.[14]

4.1 Identifying assumptions

The assumptions behind the DID approach are that, in the absence of the mining boom, residents' criminal behavior in mining municipalities would have changed similarly over time with residents' criminal behavior in control municipalities (parallel trends) (Meyer, 1995), pre-periods are not affected by treatment (no anticipation), and an individual's treatment status does not affect the potential outcome of another ("stable unit treatment value assumption", SUTVA). I check these assumptions in several ways.

First, regarding the parallel trends assumption, a violation of this assumption would imply that the observed effects might be a result of preexisting trends instead of the boom.

[13] In the robustness checks, I include time-varying individual-level controls, such as being married, having children under 18, education categories (primary, secondary, and tertiary), and economic sector which distinguishes between non-employed, primary (extraction and agriculture), secondary (manufacturing and construction), and tertiary (services, healthcare, public sector, and other). I do not include the control variables in the main specification because some of the controls could be endogenous to the mining boom (Allcott and Keniston, 2018; Pérez-Trujillo and Rodríguez-Puello, 2022; Rodríguez-Puello, 2025), becoming bad controls. The main results are robust.

[14] I classify individuals into 20-kilometer rings based on their proximity to the nearest mine, obtaining five rings in total, where those located farther away serve as controls. I obtain the following distance rings (in kilometers) from the nearest mine: $d \in \{(0, 20], (20, 40], (40, 60], (60, 80], (80, 236]\}$.

To empirically assess the validity of the "parallel trends" assumption, I estimate the following dynamic DID equation:

$$Y_{ijmt} = \alpha_i + \alpha_j + \alpha_t + \sum_{t}^{T} \beta_t \times I_t \times Treated_{imt} + \epsilon_{ijmt} \tag{2}$$

where Y_{ijmt} is equal to 1 if individual i located in grid j and in municipality m in year t is convicted of a crime. $Treated_{ijmt}$ is a binary variable that takes the value of 1 if individuals are located in mining municipalities and 0 if individuals reside in other municipalities in the county (control). The I_t's represent each year, accounting for the dynamic nature of the approach. The coefficients of interest are the β_ts, which identify the per-period difference-in-differences estimate of the effects of mining on the outcome Y_{ijmt}. I normalize β_{2003} to zero; thus, all the coefficients are interpreted as changes relative to that year. In this dynamic DID approach, the first difference is between the reference period $t = 2003$ and the period $t + x$, while the second difference is between the treated and control individuals. The β_ts for $t > 2003$ capture the dynamic effects of the treatment. On the other hand, the β_ts for $t \leq 2003$ provide a placebo or falsification test for the parallel trend assumption. In this specification, I include the same fixed effects to account for omitted variables and isolate the effect of the event.

Second, if there are spillovers to neighboring control municipalities, the SUTVA assumption would be violated, and the parameters of interest in the main model would be biased toward zero. In other words, I assume that there is no interference between units, and the individuals in the control municipalities are not affected by the treatment via spatial spillover effects (Sinclair et al., 2012). As a robustness check, I remove residents located in the four neighboring municipalities, which are most prone to spillovers.[15] Third, I assume that there are no time-varying omitted variables at the treatment level correlated with the boom and the outcomes. Specifically, I assume that individuals in treated and control locations are similar in the time-varying evolution of observed and unobserved characteristics (Von der Goltz and Barnwal, 2019). The fact that there is little to no change in the results when including the control variables supports this assumption. Moreover, although the DID design only requires that treatment and control groups exhibit the same trends (not necessarily the same levels) in the absence of treatment, one could worry that the control group does not provide an adequate counterfactual in light of the level gap. Online Appendix Table A.3 shows that individuals in the pre-boom years are close to each other in observed characteristics. More importantly, in terms of trends, Online Appendix Table A.4 shows the changes in individual characteristics between 2000 and 2003 for treated individuals compared with control individuals and the mean difference test. I do not find any economically meaningful differences in trends across groups, and only a few characteristics have p-values less than

[15] The four neighboring municipalities are Jokkmokk, Pajala, Överkalix, and Boden.

0.05. Finally, an additional concern is endogenous self-selection, where individuals may have chosen to migrate to the mining area, anticipating that the move would improve their living conditions. To address this concern, the main specification excludes individuals who moved to the treated or control locations after 2004 (migrants) (Benshaul-Tolonen et al., 2019; Jacobsen et al., 2023).

5 Results: Mining boom and crime

5.1 Main results

I begin by estimating the overall effect of local economic shocks on the different types of crimes committed by residents. Table 1 reports the DID coefficients from equation (1) for 18-29-year-old male residents. All estimations include individual, year, and grid fixed effects, do not include control variables, and are the preferred specification since some controls could be endogenous to the mining boom and criminal behavior (Allcott and Keniston, 2018). All robustness tests, including controls and different fixed effects, among others, are in subsection 5.5.

Column (1) shows the results for being convicted of any crime, column (2) property crime, column (3) violent crime, column (4) traffic crimes, and column (5) substance-related crimes. Results for the combined "any crime" outcome show a statistically insignificant effect due to the mining boom, as the reduction in property crime and the increase in substance-related crime largely offset each other. This reinforces the importance of disaggregating by crime type, as aggregate measures obscure the opposing behavioral responses documented in this paper. Given this, the remainder of the analysis focuses on disaggregated crime types.

The results suggest a negative and significant reduction in the probability of being convicted of property crime after the mining boom for treated young residents. I observe a decline of 0.66 percentage points in the probability of being convicted of property crime among treated individuals relative to their non-treated counterparts. From a baseline sample mean of 0.012, this estimate translates to a 52% drop in individuals convicted and is statistically significant at the 5% level. These declines are not confined to a single offense type: disaggregating by Criminal Code chapter shows broadly similar reductions across theft and appropriation offenses, breach-of-trust crimes such as embezzlement and misappropriation, and damage-related offenses including vandalism, with statistically significant declines in the latter two categories (Online Appendix Table A.5).[16] These findings are in line with recent micro-level evidence (e.g., Axbard et al., 2021; Street, 2025). For example, Axbard et al.

[16] Chapter 8 (Column 1) of the Criminal Code covers theft, robbery, and other unlawful appropriation offenses, including shoplifting and burglary, which are most directly linked to economic incentives and short-run opportunity costs. Chapter 10 (Column 3) includes crimes involving breach of trust, such as embezzlement and misappropriation in organizational or employment settings, while Chapter 12 (Column 5) covers damage-related offenses, including vandalism and property destruction.

(2021) finds that increased mineral wealth in South Africa leads to less crime due to changes in employment opportunities, while Street (2025) documents a 14–17.5% drop in cases filed among residents in US fracking counties.

In line with expectations, there is no significant effect on violent crimes (Column (2)), suggesting that positive local economic shocks do not directly change interpersonal violence. There is no effect on traffic-related crimes (Column (3)), and the coefficients are small and imprecise. The null for violent crime extends to gender-based violence: the mining boom has no statistically significant effect on suspected offenses in which the victim is female, consistent with the absence of a broader violent-crime response.[17]

[17] Gender-based violence is identified from the Crime Suspicion Register, which records the victim's gender. The outcome equals one if the individual is suspected of any of the following: assault against a woman, (attempted) murder of a woman, unlawful threat against a woman, stalking, molestation, sexual molestation, (attempted) rape of a woman or a person of unspecified gender, or violation of the restraining order. Such cases account for less than one percent of the young male sample, which limits statistical power.

Table 1: Impact of the mining boom on criminal behavior, 2000-2015

	(1) Any crime	(2) Property crime	(3) Violent crime	(4) Traffic crime	(5) Substance crime
Post*Treated	-0.0046	-0.0066**	0.0018	-0.0017	0.0046**
	(0.0039)	(0.0027)	(0.0016)	(0.0020)	(0.0018)
Individual FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480	230480
N	14405	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0251	0.0125	0.0060	0.0097	0.0025
Effect relative to the mean (%)	-18.25	-52.40	29.18	-17.06	181.24
R-squared	0.3445	0.2929	0.2195	0.2708	0.3759
Within R-squared	0.0000	0.0001	0.0000	0.0000	0.0000

Notes: The table reports difference-in-differences estimates of the 2004 mining boom on the probability of criminal conviction among young male residents aged 18–29, comparing Gällivare and Kiruna (treated) with other Norrbotten County municipalities (control) before (2000–2003) and after (2004–2015) the boom. Outcomes are binary indicators for conviction of: any crime, property crime (theft, robbery, fraud, embezzlement, and property damage); violent crime (violence against persons, unlawful coercion, sexual offenses, and family crimes); traffic crime (Traffic Offences Act and Maritime Offences Act); substance crime (Narcotics Act and related legislation covering possession, use, production, and distribution of narcotics, doping substances, and alcohol/tobacco products). Individuals who moved to the area after 2003 are excluded; the sample comprises 14,405 individuals and 230,480 individual-year observations. All columns include individual, year, and grid fixed effects; standard errors (in parentheses) are clustered at the residential grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Finally, there is a positive and significant increase in the probability of being convicted of a substance-related crime after the mining boom for treated young male residents (Column (4)). I observe an increase of 0.46 percentage points in the probability of being convicted of a substance crime among treated individuals relative to their non-treated counterparts. From a baseline sample mean of 0.002, this estimate translates to a 181% increase in individuals convicted and is statistically significant at the 5% level.[18] Approximately 95% of substance-related convictions in the sample fall under the Narcotics Law, which includes offenses related to possession, use, production, and distribution of narcotic substances, and the conviction

[18]The opposing signs of the estimates of property and substance crime are statistically distinguishable from each other. To test this, I stack both outcomes into a single dataset (each individual-year appears twice, once per crime type) and estimate a joint DID model that absorbs individual-by-crime-type, year-by-crime-type, and grid-by-crime-type fixed effects (Kling et al., 2007). The estimated difference between the substance and property crime coefficients is 0.0112 percentage points and is statistically significant, confirming that the two effects are statistically distinct.

data does not allow for separation between them. I provide a decomposition of substance-related crimes using suspicion data. The increase reflects changes in personal consumption rather than supply-side activity. Table A.6 decomposes narcotics-related suspicions into holding/use versus production and distribution: the overall rise is concentrated in holding and use offenses, with negligible increases in production and distribution from very low baselines.[19] This pattern is consistent with increased local demand and market activity rather than widespread entry into drug production or trafficking. On the contrary, a very small share of convictions are alcohol-, tobacco-, and doping-related offenses. The increase is not driven by alcohol-related behavior: I find no significant effect of the mining boom on convictions or suspicions under alcohol legislation, nor on intoxicated driving identified from the Crime Suspicion Register; approximately 95 percent of substance-related convictions fall under the Narcotics Law.[20]

The increase in substance-related crime is consistent with a growing body of evidence that resource booms increase risky and consumption-related behaviors, such as an increase in the demand for various goods and services, including alcohol, narcotics, and entertainment activities provided by the adult entertainment industry (e.g., Wilson, 2012; Tynan et al., 2017; Beleche and Cintina, 2018; Cunningham et al., 2020). There is no significant effect of the mining boom on young males between 30 and 39 years old (Online Appendix Table C.1).

Overall, these results reveal opposing crime responses to the mining boom among young males: a reduction in property crime consistent with the opportunity cost mechanism discussed in the introduction, and an increase in substance-related crime consistent with higher disposable income and consumption of risky goods. Together, these findings highlight that improved economic conditions do not uniformly reduce all forms of criminal behavior, and that the type of crime matters for understanding the social consequences of local economic shocks. I explore the mechanisms in more detail in Section 6.1.

Development over time. The credibility of the DID estimation hinges crucially on the parallel trends assumption. That is, the pre-2004 time trends in the outcome follow the same trend over time between the residents in the treated and control municipalities until 2004, when the mining boom started. To validate the parallel trends assumption and analyze the temporal dynamics of criminal behavior after the mining boom, I estimate a dynamic

[19] Narcotics suspicions are drawn from the Crime Suspicion Register (Brå). The decomposition into holding/use versus production and distribution provides the best available proxy for personal consumption versus market activity; minor personal-use offenses are classified as holding/use in the suspicion register. Base rates for production and distribution are very low; estimates for these sub-categories should be interpreted with caution.

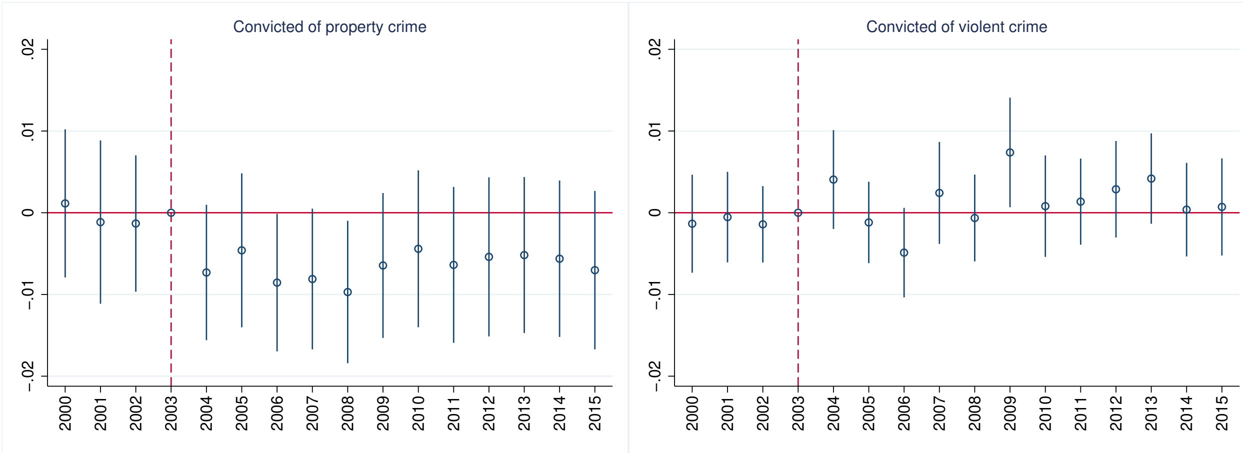
[20] Convictions under alcohol-related legislation include illegal production, possession, or sale of alcohol. Sweden's state alcohol retail monopoly (Systembolaget) limits availability by restricting the number and hours of retail outlets, imposing age verification, and setting prices above market-clearing levels. Alcohol offenses often result in fines rather than convictions; I use suspicion data as a robustness check and find consistent null results. Results are available on request.

DID (equation (2)). Figure 3 shows the dynamic treatment effect computed using the same specifications as Table 1, that is, the effect of the mining boom on the probability of being convicted of the different crime types by year. The coefficients for years 2000-2002 (before the shock) allow us to test the presence of parallel pretrends. Importantly, these coefficients are not significantly different from zero, providing evidence supporting the identifying assumption that the treated and control individuals followed the same economic trajectory before the boom. Thus, they provide support for the use of a DID empirical strategy.

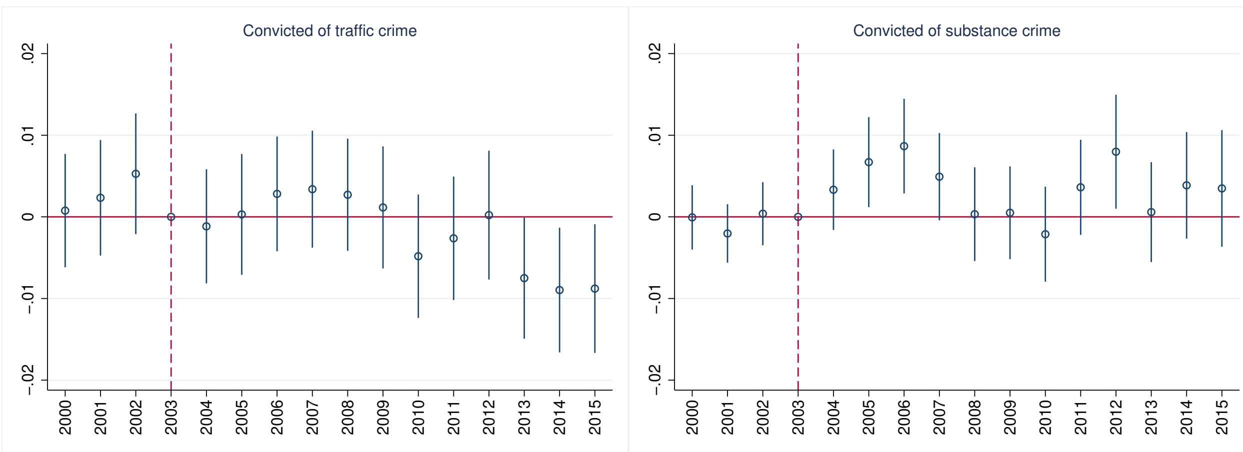
After 2004, I observe that the probability of being charged with a property crime decreases for individuals located in the mining municipalities compared to those in the control group. This effect disappears after 2009, becoming statistically insignificant, which coincides with the timing of the global financial crisis. Regarding substance crimes, I observe an increase in the probability of being charged with a substance-related crime after 2004, becoming insignificant after 2008. These findings complement the ones in Table 1 showing negative effects of the boom on property crimes and positive effects on substance crimes.[21]

[21] In addition, I estimate a dynamic DID to validate the parallel trends assumption for the spatial heterogeneity treatment (Table A.9), which is in the Online Appendix Figure B.5. Importantly, the coefficients for years 2000-2002 (before the shock), which allow us to test the presence of parallel pretrends, are not significantly different from zero. Moreover, after 2004, I observe that the probability of being charged with a property crime decreases and increases for substance crimes for individuals located within 20 kilometers of the nearest mine compared to those in the control ring.

Figure 3: Event study of the impact of the mining boom on criminal behavior, 2000-2015



(a) Property crime (b) Violent crime



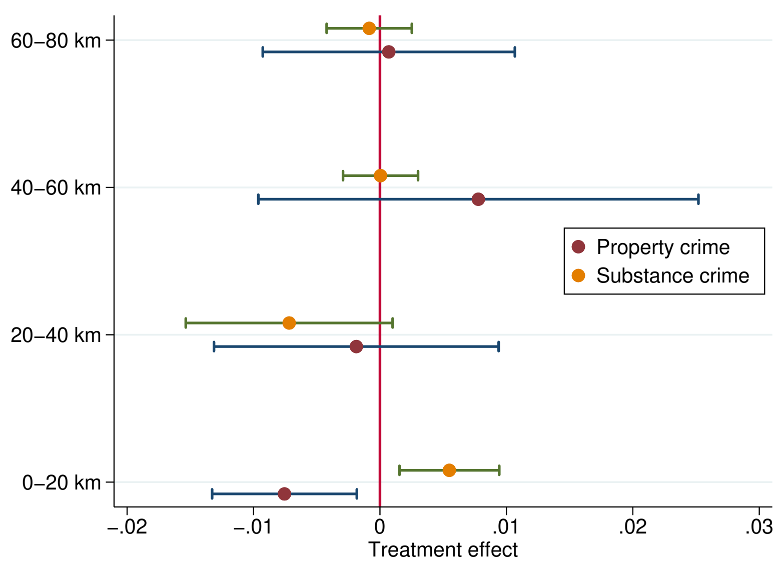
(c) Traffic crime (d) Substance crime

Notes: The figure plots year-by-year DID coefficients from a dynamic version of the main model, estimated separately for each crime type. Each coefficient measures the difference in conviction probability between treated and control residents relative to the reference year 2003 (normalized to zero). Coefficients for 2000–2002 serve as placebo tests for the parallel trends assumption. 95% confidence interval shown. Design and sample identical to Table 1. Estimations include individuals, grid, and time fixed effects. The sample excludes the migrants to the mining area. Standard errors are clustered at the grid level.

Table 1 shows no statistically significant effects of the mining boom on violent or traffic crimes, with small and imprecise coefficients. For the remainder of the analysis, I therefore focus on property crime and substance-related crime, which are the two outcomes that respond significantly to the boom and drive the opposing mechanisms documented in this paper. Estimates for violent and traffic crimes across all subsequent specifications are reported in Online Appendix A for completeness.

Spatial heterogeneity. To examine whether these effects are spatially concentrated near the mines, as expected if the mining boom is the driving force, I exploit variation in the distance from each individual's residential location to the nearest mine. I classify individuals into 20-kilometer rings and estimate the treatment effect for each ring relative to those living beyond 80 kilometers. Figure 4 reports the results.[22] I observe a large spatial heterogeneity in crime responses. There is a negative and significant reduction in the probability of being convicted of property crime after the mining boom for young male residents located within 20 kilometers of the mines. There is no significant effect for those individuals located farther away from the mines. These results provide evidence of the large spatial localization of the mining boom effects. Additionally, the increase in substance-related crimes is also concentrated among those young male residents located within 20 kilometers of the mines. The specification indicates a decline of 0.76 percentage points in the probability of being convicted of property crime among treated individuals relative to their non-treated counterparts. From a baseline sample mean of 0.012, this estimate translates to a 61% drop in individuals convicted of property crime and is statistically significant at the 1% level. This spatial gradient is consistent with Rodríguez-Puello and Rickardsson (2026), who document that the labor market effects of the mining boom spread up to 27 kilometers during the boom, and with the broader finding that the local effects of economic shocks decay sharply with distance (e.g., Feyrer et al., 2017). The alignment between the spatial reach of the labor market channel and the crime response strengthens the interpretation that improved economic opportunities are driving the observed changes in criminal behavior.

Figure 4: Treatment effects by distance to the nearest mine, 2000-2015



Notes: The figure shows DID estimates by 20-kilometer distance ring from the nearest iron ore mine (Kirunavaara/Gruvberget in Kiruna and Malmberget in Gällivare), with residents beyond 80 km as the reference category. Only property and substance crime are shown, as violent and traffic crime effects are not statistically significant (Table 1). Horizontal lines show 95% confidence intervals. Underlying coefficient estimates for all four crime types are in Online Appendix Table A.9. Design and sample identical to Table 1.

[22]The coefficient estimates, standard errors, and sample descriptives corresponding to the figure are reported in the Online Appendix Table A.9.

For the remainder of the analysis, I use the municipality-based treatment definition. The two measures are highly overlapping in practice: over 87% of Gällivare and Kiruna residents live within 20 kilometers of the nearest mine, and the average distance from the mine for Gällivare residents in the sample is approximately 8 kilometers. The municipality definition is more transparent and easier to interpret while capturing essentially the same population.

5.2 Entry into crime and recidivism

A central question for policy is whether improved economic opportunities deter entry into crime or only reduce reoffending among individuals already involved. The two margins have different welfare implications: deterrence among potential first-time offenders avoids the long-run consequences of a criminal record, while a response concentrated on re-offenders points toward post-incarceration incentives. The panel structure of the data, combined with detailed conviction information, lets me distinguish first-time from repeat offenders and separate convictions resulting in a prison sentence from those that do not.

I construct five binary outcomes that capture the nature and intensity of criminal activity (e.g., Britto et al., 2022; Grenet et al., 2024). Two split overall convictions by offender history: an indicator for being a first-time offender (the first year an individual is convicted, with no prior conviction in the panel) and an indicator for re-offense (any conviction following an earlier one), reflecting persistent or repeated criminal behavior. Three additional outcomes split convictions by whether they result in a prison sentence: convicted-without-prison, convicted-with-prison, and post-prison reoffense (any conviction in a year following a previous prison sentence). Convicted-without-prison represents the majority of convicted individuals. Convicted-with-prison serves as a proxy for more serious offenses or incapacitation. Post-prison reoffense isolates patterns of reentry into criminal activity post-incarceration.[23] Individuals with no convictions across all years constitute the reference group; the outcomes are mutually exclusive within each category.

The results reveal important heterogeneity (Table 2). The reduction in property crime convictions is concentrated among first-time offenders (Column 1), with no significant effect on re-offense (Column 2). This suggests that improved labor market conditions deter initial entry into economically motivated crime, while individuals with prior convictions are less responsive to changes in local economic conditions. This result is contrary to Britto et al. (2022), who find that crime increases for both first-time offenders and re-offenders after a job loss. For substance-related crimes, the pattern is reversed: the increase is concentrated among re-offenders (Column 4), suggesting that the boom intensifies substance-related criminal activity among individuals with pre-existing involvement rather than inducing new entry.

[23] It is important to note that this measure of recidivism is an "ever recidivist" measure within the panel window, not a rate conditional on release timing or sentence length. Therefore, I do not observe the post-prison reoffense of those individuals who are imprisoned late in the sample period, because I only observe a few years afterward.

This asymmetry between property and substance crime, where entry into property crime decreases but repeated substance offending intensifies, is consistent with the view that different crime types respond to distinct economic incentives. While improved labor market opportunities raise the opportunity cost of economically motivated crime, deterring initial engagement (Becker, 1968; Draca and Machin, 2015), substance-related offenses appear to be driven by income and consumption effects that reinforce existing behavioral patterns (Cunningham et al., 2020).

Table 2: Impact of the mining boom on first-time and repeat offenders, 2000–2015

	(1) First-time convicted Property crime	(2) Re-offense Property crime	(3) First-time convicted Substance crime	(4) Re-offense Substance crime
Post*Treated	-0.0060**	-0.0006	0.0012	0.0034**
	(0.0025)	(0.0014)	(0.0012)	(0.0014)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480
N	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0102	0.0023	0.0020	0.0006
Effect relative to the mean (%)	-58.87	-24.94	62.27	586.46
R-squared	0.1605	0.3358	0.1710	0.3654
Within R-squared	0.0001	0.0000	0.0000	0.0000

Notes: The table decomposes property and substance crime convictions by prior offending history. First-time convicted (Columns 1, 3) equals one in the first year of any conviction in the 2000–2015 panel, with no prior conviction. Re-offense (Columns 2, 4) equals one in any year following an earlier conviction. Individuals with no convictions across all years are the reference group. Design and sample identical to Table 1 (14,405 individuals, 230,480 person-years). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 corroborates the previous findings. The reduction in property crime is concentrated among those convicted without receiving a prison sentence (Column 1), with no effect on convictions resulting in prison or post-prison reoffense. Similarly, the increase in substance crimes is concentrated among convictions without prison (Column 4). Overall, these results suggest that local economic shocks, such as the mining boom, reduce new and low-severity criminal activity, while persistent criminal behavior among those with prior incarceration is less elastic to local labor market conditions. This is in line with Agan and Makowsky (2023), who finds that improved access to economic opportunities primarily reduces lower-level offending. It also complements the findings of Britto et al. (2022), where job loss increases criminal convictions predominantly among individuals without prior incarceration, suggesting that the incarceration margin represents a threshold beyond which labor market conditions have limited influence on criminal behavior.

Table 3: Impact of the mining boom on criminal behavior by incarceration status, 2000-2015

	(1) Convicted + no prison Property crime	(2) Convicted + in prison Property crime	(3) Post-prison reoffense Property crime	(4) Convicted + no prison Substance crime	(5) Convicted + in prison Substance crime	(6) Post-prison reoffense Substance crime
Post*Treated	-0.0060**	-0.0002	-0.0004	0.0026*	0.0004	0.0015
	(0.0026)	(0.0005)	(0.0009)	(0.0014)	(0.0004)	(0.0012)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480	230480	230480
N	14405	14405	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0113	0.0006	0.0006	0.0019	0.0003	0.0004
Effect relative to the mean (%)	-53.20	-24.92	-66.84	142.47	150.95	384.75
R-squared	0.2390	0.1617	0.3567	0.3215	0.1418	0.3586
Within R-squared	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Notes: The table decomposes property and substance crime convictions by incarceration status. Convicted + no prison (Columns 1, 4) = convicted without a custodial sentence in that year. Convicted + in prison (Columns 2, 5) = convicted and sentenced to prison in the same year, a proxy for more serious offenses. Post-prison reoffense (Columns 3, 6) = any conviction following a prior prison sentence within the panel (note: reoffenses of individuals imprisoned late in the panel are not fully observed). Individuals with no convictions are the reference group. Design and sample identical to Table 1 (14,405 individuals, 230,480 person-years). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Heterogeneous treatment effects using causal forests

The average treatment effects reported above mask significant heterogeneity in how the mining boom affected individual criminal behavior. To systematically characterize this heterogeneity, I use causal forest estimators via machine learning (Athey and Imbens, 2016; Wager and Athey, 2018; Athey and Imbens, 2019), which rely on data-driven sample splits to identify the relevant dimensions of treatment effect variation, rather than relying on the researcher's discretion when selecting subgroups (Britto et al., 2022). I estimate Conditional Average Treatment Effects (CATEs) for each individual based on baseline levels of observed characteristics included in the registry data (educational level, earnings, employment status, and economic sector).[24] The CATEs are average treatment effects (ATEs) conditional on a set of variables for which the treatment effects may vary. The method produces individual average treatment effects (IATEs) and group average treatment effects (GATEs). Online Appendix D provides additional details on the estimation procedure.

I find substantial heterogeneity in treatment effects. Online Appendix Figure B.6 presents the distribution of IATEs in deciles and compares the causal forest ATE to the DID estimates from Table 1. For property crime, the causal forest ATE is larger in magnitude than the DID estimate, indicating that the average effect understates the reduction experienced by the most affected individuals. For substance crime, the causal forest ATE is similar to the DID estimate and statistically significant, with a mean of 0.004. The individual effects range between -0.02 and 0.04, and the distribution of IATEs by decile ranges from a 1.3 percentage point decline in the first decile to a 0.03 percentage point increase in the last decile, indicating that while the average effect is positive, some individuals actually experience a reduction in substance crime. While these estimates have limited precision due to the relatively small sample, the systematic patterns across subgroups are informative about who is most affected by the boom.

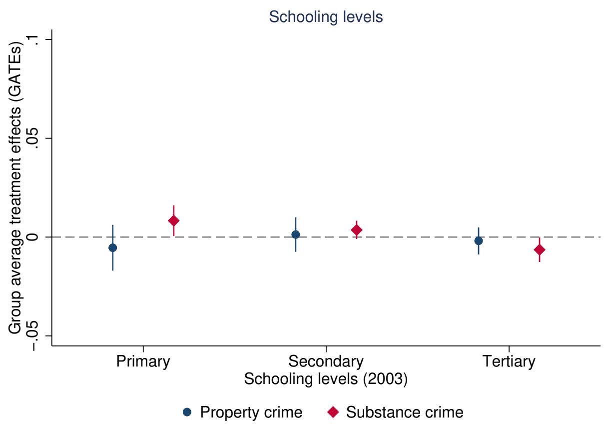
Figure 5 shows the GATEs for property and substance crimes by individual characteristics:

[24] To avoid endogenous movement across categories, individuals are classified in their education level, earnings decile, employment status, and economic sector according to their information in 2003. I analyze heterogeneity in educational levels (primary, secondary, and tertiary), across the income distribution, among employed or unemployed, and regarding employed individuals, whether those who are affected are those in sectors that are directly related to mining extraction, or whether these effects (positive or negative) are experienced in other sectors as well. This analysis is important since previous literature has found significant spillover effects of resource shocks in terms of earnings and employment to other sectors of the economy (Feyrer et al., 2017). I classify economic sectors into mining, manufacturing, construction, services, and others (including agriculture, public, and healthcare). Since the sample includes individuals as young as 18, the youngest cohorts may not have completed their education by 2003, potentially conflating low educational attainment with age. To address this concern, I replicate the education dimension of the heterogeneity analysis, restricting to individuals aged 23 or older, by which age the vast majority of Swedes have completed secondary education. The results are qualitatively unchanged, confirming that the pattern reflects educational attainment rather than age (Online Appendix Figure B.7).

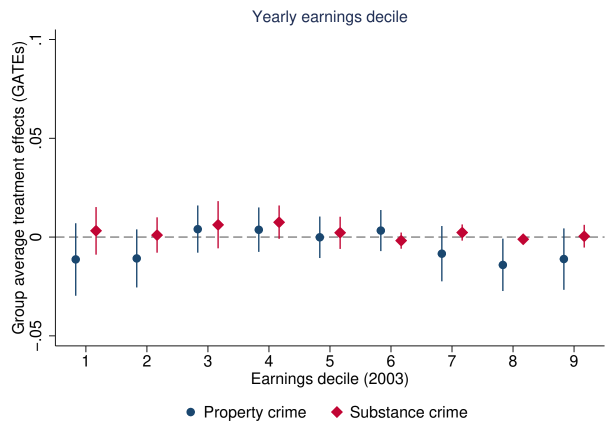
educational level, income distribution, employment status, and economic sector. On the one hand, the reduction in property crime is concentrated among young males with primary educational levels, those who are non-employed, and those in the low tail of the earnings distribution in 2003. This pattern is consistent with the economic model of crime: individuals with low educational attainment and weak labor market attachment have lower opportunity costs of crime at baseline and are therefore most responsive to improvements in economic conditions (Becker, 1968; Gould et al., 2002; Machin and Meghir, 2004). The concentration of effects among low-skilled individuals also aligns with Axbard et al. (2021), who finds that the crime-reducing effects of mining wealth in South Africa operate through employment opportunities in sectors accessible to low-skilled workers.

On the other hand, the increase in substance-related convictions is concentrated among young males with primary educational levels, employed directly in the mining sector, and in the high tail of the earnings distribution in 2003. There is a reduction in the probability of being convicted of substance crimes for young males with a tertiary educational level. The concentration among those employed in mining and in the upper part of the earnings distribution is consistent with an income effect channel, where higher disposable income increases consumption of risky goods (James and Smith, 2017; Cunningham et al., 2020), rather than an opportunity cost mechanism.

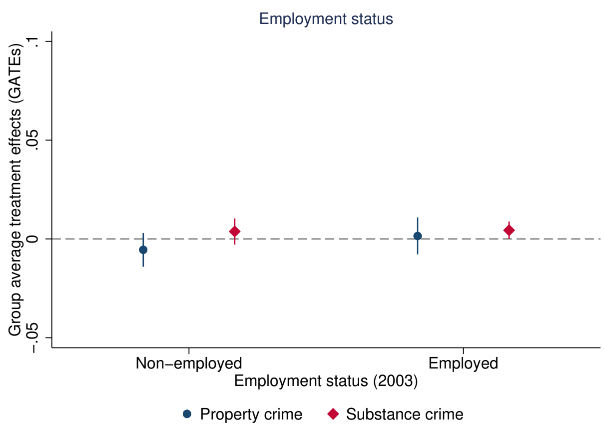
Figure 5: Group average treatment effects (GATEs) by characteristics for property and substance crimes, 2000-2015



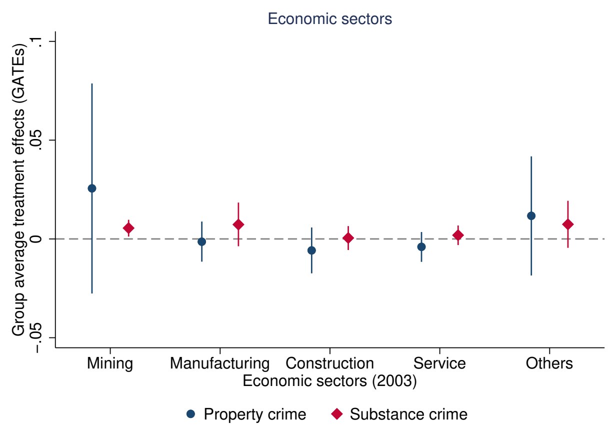
(a) Schooling levels



(b) Yearly earnings decile



(c) Employment status



(d) Economic sectors

Notes: The figure shows Group Average Treatment Effects (GATEs) for property and substance crimes from a causal forest estimator (Wager and Athey, 2018). GATEs are estimated using causal forest algorithms. Each panel plots the mean predicted treatment effect for groups defined by a pre-boom characteristic measured in 2003: schooling level (panel a), earnings decile (panel b), employment status (panel c), and economic sector (panel d). Characteristics are fixed at 2003 values to avoid post-treatment endogeneity. Bars show 95% confidence intervals. Property crime in navy (circles); substance crime in cranberry (diamonds). The individual IATE distribution is in Online Appendix Figure B.6; estimation details in Online Appendix D.

Taken together, the causal forest results reveal that the same economic shock produces opposing crime responses through distinct channels operating on different individuals. Property declines are concentrated among young males with weak baseline labor market attachment (primary education, non-employment, and the lower tail of the earnings distribution), precisely the groups for whom improved opportunities represent the largest change in the returns to legal activity. Substance increases, in contrast, are concentrated among individuals employed in the mining sector and in the upper tail of the earnings distribution, consistent with an income effect that relaxes the budget constraint for risky consumption. This separation reinforces the interpretation that property crime responds to opportunity costs while substance crime responds to income and consumption effects. Online Appendix Table A.7 confirms this directly: the statistically significant increase in substance crime is concentrated among individuals with no pre-boom property crime record, while the coefficient among prior property offenders is indistinguishable from zero, ruling out crime-type switching as a mechanism. A complementary descriptive check confirms the same pattern from the supply side: property crime declines are largest among individuals who newly enter mining employment post-2004, while substance crime rises among those already employed in mining before the boom, precisely the group that experienced large earnings gains without a change in employment status (Online Appendix Table A.8).[25]

5.4 Effects on migrants

Local economic shocks attract individuals looking for better labor market opportunities (Black et al., 2005; Komarek, 2016; Wilson, 2022). The consequences of positive economic shocks may be exploited by migrants rather than residents (Guettabi and James, 2020; Winters et al., 2021; Wilson, 2022). Moreover, as the mining sector is predominantly composed of young male individuals (a more crime-prone demographic) (James and Smith, 2017; Pérez-Trujillo and Rodríguez-Puello, 2022), improved labor market conditions may lead to a shift in population composition in mining municipalities, as a resource boom attracts workers, which could impact criminal behavior. Therefore, I describe the demographic characteristics of migrants to the mining area and empirically analyze the effects of the mining boom on the criminal behavior of young male migrants. Since migration reasons are unobserved, I define migrants as individuals who moved to Norrbotten County in 2004 or later and assume they relocated in response to improved labor market conditions.

There are notable differences in demographic characteristics between residents and migrants to the mining area (Online Appendix Table A.10).[26] First, I find that a higher share of migrants to the county are convicted of crimes, compared to residents in the mining municipalities. This is descriptive and a rough estimate, but it can be thought of as a conservative estimate of the difference in criminal propensity between groups. Second, migrants are, on average, less likely to be married and have higher educational attainment. Their employment rates and earnings are also lower compared to those of residents, particularly before 2004, suggesting more limited economic opportunities. These findings are similar to those found on migrants to US states due to fracking, who are primarily young, male, unmarried, and white (Wilson, 2022).

[25]Since sector transitions are endogenous to the boom, this exercise is descriptive rather than causal.

[26]The sample for this descriptive table includes residents and migrants between 18-39-year-old males.

While the main specification estimates the effect of the mining boom on the criminal behavior of residents, Table 4 shifts focus to young male migrants and shows the effects of the mining boom on their probability of being convicted of any crime. The table presents results from several model specifications that differ in comparison groups and estimation approaches. Columns (1)–(4) estimate conventional two-way fixed-effects regressions comparing migrants to the mining municipalities or the control municipalities to themselves before migration (the estimation includes two post-migration interaction dummies, with "pre-migration years" as the omitted category). Columns (5)–(8) contrast migrants to the mining municipalities with migrants to the control municipalities. These specifications allow me to estimate the combined impact of migration and the mining boom on individual criminal behavior.

The results indicate that migration is associated with changes in criminal behavior, but the role of the mining boom differs across crime types. Columns (1)–(4) show that, following migration, young male migrants experience an increase in property crime convictions both in mining and in control municipalities. The magnitude is slightly larger in mining municipalities, but a statistically significant increase is also present for migrants to the control areas, suggesting that the rise in property crime is largely driven by migration itself rather than by exposure to the mining boom. For violent crime, post-migration convictions increase for migrants in the control municipalities. A similar pattern emerges for substance crime: post-migration convictions decrease for migrants in the control municipalities, while no significant change is observed in mining municipalities. This suggests no meaningful differential effect attributable to mining activity. For traffic crimes, there is no evidence of systematic changes following migration in either type of municipality.

Columns (5)–(8) directly compare migrants who move to mining municipalities with those who move to non-mining municipalities, isolating the additional effect of the mining boom. These estimates show no statistically significant differential effect of mining exposure on property, violent, or traffic crime convictions among young male migrants. In contrast, column (8) reveals a positive and statistically significant effect on substance-related crime convictions: migrants relocating to mining municipalities are more likely to be convicted of substance crimes relative to migrants settling in control municipalities. While the baseline probability of substance convictions among migrants is low, the relative effect is sizable, suggesting that mining-driven local conditions amplify substance-related criminal activity among migrants.

Table 4: Impact of the mining boom on criminal behavior of migrants, 2000-2015

	(1) Property crime	(2) Violent crime	(3) Traffic crime	(4) Substance crime	(5) Property crime	(6) Violent crime	(7) Traffic crime	(8) Substance crime
Post*Migrants (Mining mun.)	0.0107**	0.0041	-0.0084	-0.0006	0.0017	-0.0039	-0.0081	0.0105*
	(0.0052)	(0.0033)	(0.0052)	(0.0051)	(0.0058)	(0.0039)	(0.0058)	(0.0057)
Post*Migrants (Control mun.)	0.0093***	0.0070***	0.0010	-0.0078***				
	(0.0033)	(0.0022)	(0.0028)	(0.0029)				
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nxt	155753	155753	155753	155753	50603	50603	50603	50603
N	9735	9735	9735	9735	3163	3163	3163	3163
Mean dep. var (2000-03)	0.0616	0.0205	0.0446	0.0466	0.0169	0.0063	0.0108	0.0058
Effect relative to the mean, Treated (%)	17.32	19.82	-18.78	-1.20	9.92	-61.02	-74.81	181.32
Effect relative to the mean, Control mun. (%)	15.12	34.27	2.26	-16.69	0.00			
R-squared	0.4892	0.3494	0.4901	0.5258	0.4455	0.3270	0.3891	0.4634
Within R-squared	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001

Notes: The table estimates the mining boom's effect on criminal behavior among young male migrants aged 18–29. Migrants are individuals who relocated their registered municipality of residence to Norrbotten County in 2004 or later, assumed to have moved in response to improved labor market conditions; they are excluded from the main resident sample (Table 1). Columns (1)–(4) compare migrants against themselves before migration, separately for those who moved to mining municipalities (Gällivare/Kiruna) versus control municipalities (N = 9,735 individuals, 155,753 person-years); pre-migration years are the reference. Columns (5)–(8) directly compare mining-destination migrants with control-destination migrants, isolating the additional effect of mining exposure conditional on migration (N = 3,163 individuals, 50,603 person-years). Crime definitions are in Table 1. All columns include individual, year, and grid fixed effects; standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Overall, the mining boom does not broadly increase criminal behavior among migrants: effects are concentrated in substance-related offenses, consistent with income and local-market mechanisms, while other crime types show no differential impact relative to control-area migrants. This complements Street (2025), who finds that aggregate crime increases in US fracking areas are driven by in-migration rather than by pre-existing residents, a pattern our individual-level results corroborate.

5.5 Robustness checks

The estimated impacts of the mining boom on criminal behavior are robust to various alternative specifications and robustness checks.

Data-driven rings. The spatial analysis in Section 5.1 classifies individuals into 20-kilometer rings to examine distance decay. A natural concern is whether the results are sensitive to this choice of cutoff. According to Butts (2023), the wrong choice of cutoff biases the results, while the correct identification of the cutoff enables an enhanced understanding of the spatial propagation of the treatment effects. I use an alternative nonparametric estimator that provides a more complete picture of how the shock affects units at different distances, proposed by Butts (2023).[27] It estimates a curve that represents the effect as a function of distance by using many rings. In addition, it selects the rings in a data-driven procedure, eliminating the need to specify a cutoff where the treatment effects become zero to estimate the average treatment effect (Cattaneo et al., 2019), thereby avoiding potential specification searching (Andrews and Kasy, 2019). Using this method, I obtain ten rings grouping the young males sample according to their distance to the nearest mine. The results, in Online Appendix Table A.11, show that the findings are robust to this empirical strategy.

Travel time treatment. Next, in the characterization of the spatial decay using distance to the nearest mine, straight-line distance may mischaracterize actual accessibility in northern Sweden, where road infrastructure is sparse, and routes between settlements are often circuitous. As a robustness check, I redefine treatment using travel time by car, measured with OpenStreetMap data via the Open Source Routing Machine (OSRM). I classify individuals into 20-minute rings, obtaining a total of five rings, where individuals located farther away serve as controls. Online Appendix Table A.12 reports the results. The estimated effects for being convicted of property and substance-related crime are virtually unchanged, compared to those of Table 1, confirming that the findings are robust to this alternative treatment definition. Specifically, there is a negative and significant reduction in the probability of being convicted of property crime after the mining boom for young male residents located within 20 minutes by car of the mines, and an increase in substance crimes for the same treated individuals.

[27] According to Butts (2023), this method is similar to using the distance to the nearest mine as a continuous measure to estimate the "dosage-response" function proposed by Callaway et al. (2024) in the difference-in-differences approach with continuous treatment.

Synthetic control. Although the main analysis is at the individual level, all treatment variation comes from only two municipalities. The synthetic control method (Abadie et al., 2010) provides aggregate-level validation of the individual results, constructing a municipality-level counterfactual that does not rely on the parallel trends assumption. I follow Galiani and Quistorff (2017) and pool Gällivare and Kiruna into a single composite treated unit, constructing a weighted combination of control municipalities that minimizes pre-treatment prediction error over 1996–2003.[28] Online Appendix Figure B.8 shows the synthetic control tracks pre-boom crime trends closely for both property crime (panel a) and substance crime (panel b). The post-boom divergence mirrors the main DiD estimates: property crime falls below the counterfactual from 2004 onward while substance crime rises above it, corroborating the causal interpretation at the aggregate level.

Additional robustness checks. Online Appendix Table A.13 reports seven robustness exercises for the main results of young males: (1) baseline, (2) treatment defined by 2003 municipality of residence, (3) inclusion of time-varying controls, (4) inclusion of migrants, (5) balanced panel, (6) excluding four neighboring spillover-prone municipalities, (7) municipality FE in place of grid FE. Coefficients are stable across all specifications.

6 Contextualization and mechanisms

6.1 Why does the mining boom affect crime?

Generally, the estimates are reduced-form effects that encompass multiple potential mechanisms. However, I claim that the estimates are consistent with an opportunity cost mechanism from an improvement in labor market conditions.[29] The results reveal two opposing crime responses that operate through distinct channels, which I discuss in turn. In this section, I use the same empirical design described in Section 4 (using the variables capturing these mechanisms as outcomes) to explore the first-stage effect for the mechanisms via which a mining boom might affect criminal behavior, even though I cannot definitively distinguish across them or rule out the possibility that there are other intermediating variables at work.[30]

[28] The donor pool excludes municipalities with a small mine, those in the Stockholm, Gothenburg, and Malmö metropolitan regions, and those with high steel industry employment, the same exclusions as the main analysis.

[29] While this paper builds on the economic model of crime, sociological theories emphasize complementary channels, such as social integration, cultural conflict, social inequality, and a breakdown of social control (Cohen et al., 1981; Pratt and Cullen, 2005). Some of these channels, for instance, the routine activity theory, which contends that criminals act when there is a convergence of likely offenders, suitable targets, and an absence of capable guardians against crime (Cohen and Felson, 1979), may overlap with the economic mechanisms discussed here, particularly regarding crime prevention capacity and population composition.

[30] Another approach to evaluate mechanisms in the literature is to include the variable on the right-hand side as a control to see how the main treatment effect changes and test mediation or partial channeling.

6.1.1 Property crime: the opportunity cost channel

Labor market opportunity cost. If the mining boom improved local labor market conditions, it would raise the returns to legal work and shift the cost-benefit calculation away from criminal activity (Becker, 1968; Ehrlich, 1973).[31] As documented in Section 2, resource booms of this type raise local employment and earnings significantly (Black et al., 2005; Allcott and Keniston, 2018), including in the Swedish case (Tano et al., 2016; Rodríguez-Puello and Rickardsson, 2026), making labor market conditions a natural channel through which the boom may have affected crime.[32] At the same time, higher earnings also make potential victims wealthier, raising the expected payoff to property crime, the so-called rapacity effect (Draca and Machin, 2015). The evidence below is inconsistent with this channel dominating.

I start by examining how the mining boom affects the labor market conditions of young male residents in the mining municipalities and discuss its relative importance in explaining the changes in crime as a result of the resource shock (Table 5). Online Appendix Figure B.9 provides a visual overview of the labor market trends in mining versus non-mining municipalities over the study period. Columns (1), (2), (3), (4), and (5) show the results for disposable income, labor income (earnings), labor income conditional on being employed, employment overall, and employment in the mining sector, respectively.[33] Consistent with previous work, the mining boom raises labor market opportunities on both the extensive and intensive margins. Specifically, the mining boom has a positive effect on the labor market conditions of young male residents in the mining municipalities, with a significant increase in disposable and labor income, and the probability of being employed, especially directly in the mining sector. Yearly disposable income increases by 17,923 SEK for treated young male residents after the mining boom compared with control residents. This represents a 17% increase from the baseline mean. The observed increase in labor income is higher. While

Nevertheless, the main concern why I do not apply it is that mediators (e.g., earnings) are bad controls and bias the treatment effect.

[31] Theoretically, individuals are rational economic agents that choose between legal work and criminal activity by comparing the legal work wage in the labor market and the expected payoff to crime (which depends on the expected gain from crime minus the cost, which is the product of the probability of being caught and the associated punishment), choosing crime whenever the former exceeds the legal wage (Becker, 1968; Ehrlich, 1973). Therefore, individuals with higher wages or better employment opportunities experience an increase in the opportunity cost of engaging in criminal activity, reducing local crime levels.

[32] The link between labor market conditions and crime has also been explored in the literature (e.g., Raphael and Winter-Ebmer, 2001; Edmark, 2005; Öster and Agell, 2007; Fougère et al., 2009; Nordin and Almén, 2017; Dix-Carneiro et al., 2018; Ferraz et al., 2022).

[33] Disposable income is the sum of all incomes, including other benefits (e.g., child allowances, social benefits, and housing benefits) minus final tax. All income variables are adjusted to real values with the base year 2000 using the national CPI. To avoid typical problems of zeros in the outcome variables (Chen and Roth, 2024; Mullahy and Norton, 2024), I measure income in levels. Therefore, the coefficients can be interpreted as the effect on income as measured directly in 1000 Swedish krona (in 1000 SEK).

there is no clear effect on employment due to different effects for different economic sectors, a substantial increase is observed in the probability of being employed in the mining sector.[34] These labor market improvements raise the returns to legal activity, shifting the cost-benefit calculation away from property crime (Draca and Machin, 2015; Edmark, 2005; Axbard et al., 2021).[35]

Table 5: Mechanisms: impact of the mining boom on labor market outcomes, 2000-2015

	(1) Disposable income	(2) Labor income	(3) Lab. inc. employed	(4) Employment	(5) Employment mining
Post*Treated	17.9232***	28.3618***	26.1973***	0.0186**	0.0869***
	(1.4464)	(2.2079)	(2.2928)	(0.0094)	(0.0068)
Individual FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes	Yes
Nxt	230480	230480	152308	230480	230480
N	14405	14405	9519	14405	14405
Mean dep. var (2000-03)	107.6874	114.5652	172.4796	0.6225	0.0219
Effect relative to the mean (%)	16.64	24.76	15.19	2.99	396.29
R-squared	0.6161	0.7432	0.7152	0.5618	0.6831
Within R-squared	0.0007	0.0025	0.0024	0.0000	0.0083

Notes: The table estimates the mining boom's effect on labor market outcomes using the same design and sample as Table 1. Disposable income (Col. 1) and labor income (Cols 2-3) are in 1,000 SEK, adjusted to real 2000 values using the national CPI. Labor income, employed (Col. 3), is labor income conditional on employment (9,519 individuals). Employment (Col. 4) and mining employment (Col. 5) are binary indicators. Standard errors (in parentheses) clustered at the grid level. Data from administrative registers and LISA, Statistics Sweden. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To examine the wage rate component of the annual earnings increase, I supplement the LISA annual earnings measure with employer-reported full-time equivalent monthly salaries from the Lönestrukturstatistik (LSS), Statistics Sweden's Structure of Earnings Survey.[36]

[34] Rodríguez-Puello and Rickardsson (2026) find that the mining boom in Sweden increased employment in mining and manufacturing, while there is a reduction in service employment due to high competition for workers.

[35] This operates through both higher returns to legal work and a direct time-use effect: employment occupies hours that would otherwise be available for criminal activity, reinforcing the opportunity cost mechanism (Cohen and Felson, 1979; Lochner, 2004).

[36] The Lönestrukturstatistik is an annual survey conducted by SCB covering the entire Swedish economy. It is based on a stratified sample of enterprises with ten or more employees, with near-complete enumeration of the public sector. Employers report wages and salary components directly; the monthly wage measure used here is standardized to full-time equivalents, making it interpretable as the wage rate independently of individual working-time arrangements. The matched sample linked to my existing administrative panel covers 3,251 individuals and 52,020 person-year observations over 2000–2015. Because the survey targets larger enterprises, coverage in small northern municipalities is sparser than in LISA, and results should be

Because the LSS wage measure is already standardized to full-time equivalents, it captures the wage rate per se rather than total compensation. The mining boom raises the monthly wage rate by approximately 3.4 percent for young male individuals in treated municipalities (Online Appendix Table A.14). The annual earnings effect reported in Table 5 is larger, mainly because it reflects both the monthly wage rate and the number of months employed during the year.

To benchmark the magnitude of these effects, I compute an implied elasticity of property crime with respect to earnings by dividing the estimated effect on property crime by the 24.8 percent increase in labor income (Column 2, Table 5), yielding an elasticity of approximately -2. That is, a one percent increase in earnings is associated with a two percent decrease in property crime conviction probability. This is at the upper end of existing estimates in the literature, which find implied elasticities roughly between -1 and -2.5 (e.g., Gould et al., 2002; Machin and Meghir, 2004; Agan and Makowsky, 2023), and substantially larger than the -0.58 estimated by Britto et al. (2022) using job loss as a shock. It is important to note that this elasticity should not be interpreted as causal, as the mining boom may affect criminal behavior through channels other than earnings, including crime prevention capacity, migration, and income inequality, which I discuss below.

Income inequality. According to the economics literature on crime, there are rational incentives to commit crimes when there are lower-income people near high-income people in a community (Deller and Deller, 2010), and the economic gains of a mining boom may be concentrated among specific population groups, such as extraction workers rather than other residents (Hardy and Kelsey, 2015). There is empirical literature linking resource booms with income inequality (e.g., Reeson et al., 2012; Loayza et al., 2013) and income inequality with crime and violence (e.g., Kelly, 2000; Bourguignon et al., 2003; Neumayer, 2005). Therefore, combining both pieces of evidence, a mining boom that increases local income inequality may indirectly generate incentives to commit crime. While I do not observe property crime increases due to the mining boom, only increases in substance-related crimes, it is important to examine this mechanism to discard its role in the main results.

Measuring income inequality at the individual level is a challenge. I examine the effect of the mining boom on the probability of moving into (or out of) the top and bottom of the income distribution. Specifically, using the labor income in 2003, I classify individuals by year and municipality of residence into terciles, and use binary outcomes equal to one if the individual is in the first tercile (bottom of the income distribution) or the third tercile (top of the income distribution). The results in Columns (1) and (2) of Table 6 show that the mining boom significantly reduces the probability of being in the bottom earning tercile (Column (1), significant at the 5% level) and weakly increases the probability of being in the top earning tercile (Column (2), significant at the 10% level). These results suggest that the boom shifts

interpreted with this limitation in mind.

the income distribution upward, benefiting individuals at both ends: lifting some out of the lowest-income group while moving others into the highest-income group. Importantly, the stronger and more precisely estimated effect at the bottom of the distribution indicates that the income gains from the boom are not solely concentrated among top earners. This pattern is more consistent with a general improvement in economic conditions than with a widening of income inequality. Therefore, there is no compelling evidence suggesting that income inequality played a role in the changes in criminal behavior. This result is contrary to James and Smith (2017), who find descriptive evidence for this mechanism in the case of the impact of an energy boom on regional crime in the United States, where the resource shock increased crime rates in shale-rich counties, and this coincided with a rise in income inequality.

Table 6: Mechanisms: impact of the mining boom on different mechanisms, 2000-2015

	(1) Bottom earning tercile	(2) Top earning tercile	(3) Police occupation	(4) Police industry
Post*Treated	-0.0108**	0.0071*	0.0033	0.0026
	(0.0054)	(0.0040)	(0.0023)	(0.0021)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	124267	124267	187619	230480
N	7767	7767	11726	14405
Mean dep. var (2000-03)	0.4136	0.2197	0.0019	0.0052
Effect relative to the mean (%)	-2.62	3.23	170.34	50.07
R-squared	0.9091	0.9176	0.5546	0.5200
Within R-squared	0.0001	0.0001	0.0001	0.0000

Notes: The table estimates the mining boom's effect on different mechanisms using the same design and sample as Table 1. Bottom/top earning tercile (Cols 1–2) indicate whether the individual falls in the bottom or top third of the annual labor income distribution within municipality and year (7,767 individuals with positive income). Police occupation (Col. 3) and police industry (Col. 4) are binary indicators for employment in police-related occupations or the justice sector (11,726 individuals). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers and LISA, Statistics Sweden. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

6.1.2 Substance crime: income and occupational stress channels

The increase in substance-related offenses follows a different logic from the property crime decline. As documented in Sections 5.2 and 5.3, the rise is concentrated among repeat offenders and young males employed directly in the mining sector, not among first-time offenders or individuals in the lower tail of the earnings distribution. This pattern points away from the opportunity cost channel and toward income and consumption effects operating among individuals already embedded in substance use.

Four mechanisms could plausibly explain this pattern: (i) an income effect, meaning that higher earnings relax the budget constraint for drug and alcohol purchases; (ii) workplace culture specific to male-dominated extractive industries; (iii) occupational stress from the rapid expansion of operations; and (iv) increased employer drug testing and occupational health screening, which would raise the detection rate of substance offenses without any behavioral change. Mechanisms (i) and (iii) are addressed through regression evidence and newspaper text analysis below. Mechanism (ii) is examined through the pattern of heterogeneity across crime types and occupational groups, alongside the existing literature on extractive industry cultures. Mechanism (iv) is addressed through institutional documentation of LKAB's testing policy.

Income channel. The most coherent of the four is the income effect. Higher earnings raise the purchasing power of individuals already inclined toward substance use, enabling greater consumption of drugs and alcohol as normal or even luxury goods (James and Smith, 2017; Cunningham et al., 2020). This is the demand-side counterpart to the opportunity-cost argument: the same income gain that makes property crime less attractive also relaxes the budget constraint on risky consumption. Consistent with this, a large literature documents that positive income shocks can increase engagement in addictive or health-risky behaviors even as they reduce economically motivated crime (Draca and Machin, 2015; Cunningham et al., 2020). The narcotics offenses decomposition by type above is consistent with this channel: the increase in substance crime is concentrated in holding and use offenses rather than in distribution, pointing to personal consumption rather than supply-side market entry. The first-stage evidence in Table 5 directly supports this channel. The causal forest results in Figure 5 reinforce this reading: the increase in substance crime is concentrated among individuals in the upper earnings distribution in 2003, those who gained most in absolute income terms from the boom.

Since the causal forest conditions on 2003 characteristics, this concentration among high earners points to pre-boom mining workers, those who gained most in absolute terms, as the primary margin of the income effect. The sector transition analysis in Section D corroborates this: the substance crime increase is concentrated among treated residents who did not newly enter mining post-2004, that is, those already employed in the sector before the boom. New mining entrants show no significant increase in substance crime. This suggests that the income effect on substance consumption operates primarily through the earnings gains of established mining workers rather than through the initial employment margin of boom entrants. This income-driven pattern is not specific to mining, as industries combining physically demanding work with above-average wages, including construction, oil and gas extraction, and transportation, systematically exhibit higher substance use rates than white-collar occupations (Frone, 2006; Bush and Lipari, 2015), consistent with a compensating-differential channel in which higher earnings expand the budget available for risky consumption.

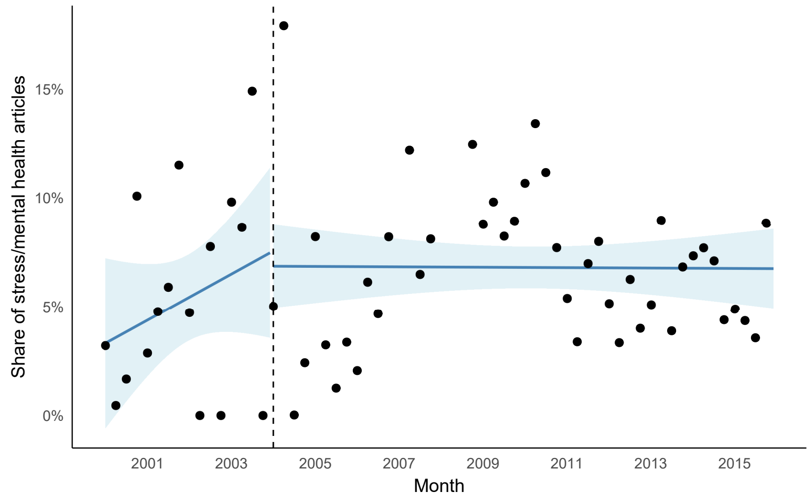
Workplace culture. A second channel, distinct from pure income effects, operates through the social norms and behavioral culture of male-dominated extractive industries, where intense physical labor, long shifts, and social isolation create conditions for collective substance use in off-hours (Shandro et al., 2011; Arratia-Solar and Paredes, 2023). Higher earnings facilitate this pattern by relaxing financial constraints. While I cannot isolate the workplace culture channel from the income effect empirically, since both predict a substance crime increase concentrated among mining workers, the evidence is at a minimum consistent with workplace culture operating as an amplifying factor alongside the income channel.

Occupational stress. A second channel operates through occupational stress. The mining boom shifted the sectoral composition of employment in treated municipalities. Table 5, Column (5) shows that the probability of being employed in the mining sector increased among treated young male residents, while Rodríguez-Puello and Rickardsson (2026) document that the boom raised employment in mining and manufacturing while reducing service sector employment as competition for workers intensified. Mining and extractive industries are among the most physically demanding and hazardous occupational environments, characterized by elevated injury risk, irregular shift schedules, and the psychological pressures of high-intensity underground operations (Shandro et al., 2011; Arratia-Solar and Paredes, 2023). These are well-documented conditions that can increase substance use as a coping response, even among individuals with rising incomes.

I provide suggestive evidence on the occupational stress channel using a text analysis of Swedish newspaper coverage of LKAB and interrupted time series (ITS) models over 2000–2015. LKAB is the main iron ore producer in Sweden and operates the large-scale mines in Kiruna and Gällivare. Full details are in Online Appendix F; I summarize the main findings here.

I construct the monthly share of LKAB-related articles mentioning stress or mental health terms and estimate an ITS model with a structural break at January 2004. Figure 6 plots the series alongside the fitted values. The share of stress-related articles is low throughout the period, averaging approximately 3 percent in 2000 and around 7 percent by late 2003. At the onset of the boom, there is no significant level shift: the estimated immediate level shift is $\hat{\beta}_1 = -0.007$, indicating no detectable change in the share of articles mentioning occupational stress or mental health. The pre-boom trend ($\hat{\beta}_2 = 0.001$) and post-boom slope change ($\hat{\beta}_3 = -0.001$) are both statistically insignificant, providing no evidence of systematic trends in stress-related coverage before or after the mining boom. The null result is robust to weighting by article count (Online Appendix Table F.4) and to dropping months with fewer than five total LKAB articles, a restriction that addresses concerns about inflated share variance in sparse pre-boom months. In the latter sample, the estimated level shift is $\hat{\beta}_1 = 0.008$, and neither the pre-boom trend nor the post-boom slope change is statistically significant (Online Appendix Table F.5). I interpret this as providing no support for an occupational stress channel.

Figure 6: Interrupted time series of stress/mental health newspaper articles, 2000–2015



Notes: The figure plots a quarterly binscatter of the monthly share of LKAB*-related newspaper articles mentioning stress or mental health terms (keyword dictionary in Online Appendix Table F.3) over 2000–2015. LKAB is Sweden's state-owned iron ore company, operating the mines in Kiruna and Gällivare. Fitted lines show estimated pre- and post-boom linear trends from an interrupted time series (ITS) regression with a structural break at January 2004 (vertical dashed line). Articles are identified using the newspaper archive *Retriever Mediearkivet*. Following the literature, we search for articles containing the case-insensitive string "LKAB*", where the asterisk is used as a wildcard. Newey-West standard errors (lag = 12). None of the ITS coefficients is statistically significant, providing no evidence of a systematic increase in occupational stress coverage following the boom. Coefficient estimates in Online Appendix Table F.4; full details in Online Appendix F.

Two additional results corroborate this reading. First, I find a null result on long-term sick leave: the mining boom generated no statistically significant increase in sick-leave spells exceeding two weeks or sickness-benefit payments among young male residents in treated municipalities.[37] Second, I provide a complementary test using employer-reported data on contracted work scope, a measure of working time as a share of full-time equivalents that incorporates overtime adjustments. If the mining boom increased work intensity through extended or non-standard hours, one would expect the contracted work scope to rise for workers in treated municipalities post-boom. The estimated effect is negative and statistically significant (Online Appendix Table A.14), indicating that contracted work scope, if anything, declined for young males in treated municipalities during the boom period. This direction is inconsistent with a story in which the expansion imposed more intensive work schedules on

[37]This is motivated by Shandro et al. (2011), who finds increases in pregnancies, sexually transmitted infections, and mine-related injuries during booming mine activities. While I do not observe detailed health outcomes such as hospital admissions or mental health diagnoses, I explore this margin using employer-paid sick leave spells exceeding two weeks, which capture more serious or prolonged health episodes available in the administrative data. Although sick leave of more than two weeks is an imperfect proxy for health and mental health, and does not capture day-by-day sickness related to substance use, prior work has used extended sickness absence as an indicator of underlying health shocks.

affected workers. Taken together, if the boom had induced widespread occupational stress or health deterioration, one would expect the share of stress-related media coverage, the incidence of severe health-related absences, and the overtime work to increase. Together, these results substantially narrow the scope for occupational stress as a driver of the increase in substance crime, even though the sectoral composition of employment shifted toward more physically demanding industries.

Detection channel. The concern that employer drug testing mechanically inflates recorded substance convictions without any behavioral change is substantially limited by the institutional structure of LKAB's testing policy. LKAB's published workplace policy documents that a positive drug test result leads to revocation of the individual's site access permit, not to a police report or criminal referral; the police are contacted only when a vehicle operator tests positive in a breath alcohol test, which constitutes a traffic offense (LKAB, 2023). Since the conviction records used in this paper require police involvement and prosecution, internal LKAB testing cannot mechanically generate entries in the outcome variable. This institutional structure reflects Swedish labor law more broadly, as employer drug testing is legally restricted to safety-sensitive roles, and a positive test result is routed to occupational health assessment and rehabilitation rather than criminal proceedings. This applies to both direct LKAB employees and contracted workers operating on LKAB's sites. For the detection channel to explain the substance crime results, police would need to have independently increased enforcement in mining municipalities after 2004, which is a hypothesis addressed and rejected in the police capacity analysis below (Section 6.1.3).

6.1.3 Government's crime prevention capacity

Changes in policing represent a competing explanation for both results. Becker (1968) highlights that the probability of detection is an important factor to consider when examining factors influencing an individual's decision to commit a crime. Increasing the probability of being caught and/or the resulting punishment may reduce crime according to theory. Previous literature has shown that crime decreases when there is an increase in police presence (Di Tella and Schargrodsky, 2004; Machin and Marie, 2011). Therefore, a concern in interpreting the main results is that the changes in crime may be due to improvements in the government's crime prevention capacity. If the mining boom generated additional local tax revenue that funded police resources, increased police presence could explain the property crime decline through a deterrence channel rather than an opportunity cost channel (Foley, 2011; Axbard et al., 2021). Conversely, an expansion of police activity could mechanically increase recorded substance offenses without any behavioral change, by raising the probability that existing substance use is detected and prosecuted.

As an approximation to this mechanism and as a proxy of the government's crime prevention capacity, I test for changes in the police force by examining the effect of the mining boom on the probability of young male residents becoming a police worker (Columns

(3) and (4) Table 6).[38] I find no significant change in police employment or resources in the mining municipalities following the boom. This null is expected in the Swedish institutional context: police resources are funded by the state alone, not by the state and local authorities as it is in other countries (Lindström, 2015), and it does not depend on the crime level or economic conditions in each municipality. A local mining boom, therefore, cannot generate additional police capacity, regardless of its fiscal effects. The existing literature confirms that policing and detection do affect crime when they are meaningfully increased. For instance, Anker et al. (2021) finds that expanding DNA database registration reduces recidivism by around 40%, and Kirchmaier et al. (2020) finds that a targeted policing operation reduced metal theft by around 35%. However, neither condition held here. Consistent with this, James and Smith (2017) and Axbard et al. (2021) find that changes in fracking activities and mining value did not affect police operations, ruling out policing as a driving mechanism in similar settings.

Overall, the evidence points to two distinct behavioral channels operating simultaneously. The reduction in property crime is consistent with improved labor market conditions, raising the opportunity cost of economically motivated criminal activity (Becker, 1968). The increase in substance-related crime is consistent with an income and consumption channel: raising earnings relaxes the budget constraint for drug purchases, concentrated among individuals already involved in substance use and directly employed in the mining sector. The occupational stress channel receives no support from the newspaper text analysis. Changes in policing do not account for either result. Selective migration and income inequality do not drive the results. Disentangling the remaining channels further would require data on employer testing policies and workplace health records not available in the current analysis.

6.2 Social cost effects

Online Appendix Table A.15 translates the main crime effect estimates into estimates of the effect of the mining boom on social costs of crime for young males, following the methodology of Alsan et al. (2025). Specifically, I calculate the total unit cost for each crime category, which includes the costs for anticipation (e.g., defensive expenditure), consequence (e.g., physical and emotional harm), and response (e.g., police costs) to the crime, as reported in Heeks et al. (2018). I multiply the estimated coefficients of Table 1 by the cost of crime and add it

[38] There are two ways of classifying residents as police workers using data from Statistics Sweden. Since neither of them is a perfect classification, I use both to compare the results. I classify as police those individuals working in the security sector using the Swedish Standard Industrial (SNI) Classification from 2007, specifically classified in the codes 74900, 80100, 80200, 80300, and 84240. I use this data because they are available for the whole period of analysis. As a comparison, I use data from the "Swedish Occupational Register with Statistics" (Statistics Sweden) for the period 2001-2015. The data are available only after 2001, and those for the years 2014 and 2015 are not comparable. As police officers, I consider patrol officers, criminal investigators, and community police officers (Lindström, 2015). The correlation of the police per capita variables for the period 2001-2013 among the two measures is 90%.

by the number of treated young male individuals over the 12 post-boom years (2004–2015), obtaining the aggregate social cost effect (total welfare implication) of the mining boom.

The cost estimates draw on Heeks et al. (2018), who construct unit costs for each crime category using three distinct methods. Anticipation costs are derived from survey data on defensive expenditures, including spending by potential victims on security measures, locks, alarms, and insurance premiums. Consequence costs combine willingness-to-pay estimates from victim surveys, capturing physical injury, psychological distress, and quality-of-life losses. Response costs are computed from administrative records on police investigation, prosecution, court proceedings, and incarceration per crime type. A natural concern is whether these UK-based estimates transfer to the Swedish context. No analogous comprehensive cost study exists for Sweden, but the United Kingdom is a closer institutional comparator than the United States: both are high-income welfare states with broadly comparable criminal justice structures, crime category definitions, and income levels. The estimates should therefore be read as order-of-magnitude welfare calculations rather than precise accounting, because differences in incarceration rates, healthcare costs, and victim compensation schemes introduce measurement error of uncertain direction.

The results show a mixed picture. Consistent with the regression estimates, reductions in property crimes translate into sizable social savings during the boom. These savings are non-trivial for a municipality of this size, even if modest relative to national figures. However, other categories reveal offsetting costs. The effects on violent and traffic crimes are statistically insignificant. Substance crimes increase significantly during the boom, generating social losses that partially offset the property crime gains. This pattern mirrors the conviction results and suggests that mining-driven shocks have unintended consequences for illicit substance use. Taken together, reductions in property crime generate SEK 17.8 million in social savings, partly offset by SEK 14.1 million in additional costs from the substance crime increase, for a net crime-related gain of approximately SEK 3.7 million over the 2004–2015 boom period.

Two caveats are necessary before drawing policy conclusions from these estimates. First, the social cost figures capture only the crime-related welfare consequences of the boom. The boom also raised labor income by approximately SEK 28,000 per treated young male resident per year, generating aggregate income gains that substantially exceed the crime-related effects in magnitude. The net social gain from the boom is therefore substantially larger than the crime cost accounting suggests, and the crime-specific near break-even result should not be read as evidence that the boom was welfare-neutral overall. Second, the social cost of substance crime is difficult to measure precisely because it encompasses health costs, productivity losses, and criminal justice expenditures that extend well beyond the boom period. The SEK 14.1 million figure likely understates the long-run social burden of increased substance convictions, particularly if addiction persists after the boom ends. These limitations suggest that the social cost calculation is best understood as a lower bound on the full welfare consequences of the substance crime increase, and a lower bound on the net benefits of the property crime reduction.

7 Conclusions

This paper asks whether a positive local economic shock substitutes for or complements crime. The 2004 iron-ore boom in northern Sweden, combined with geocoded administrative registers on all criminal convictions and demographics in Sweden from 2000 to 2015, allows a clean answer: it does both. Among young male residents aged 18–29 in mining municipalities, property crime convictions decline by 52 percent and substance-related convictions rise by 181 percent, with no effect on violent or traffic crimes. These effects are driven by existing residents in the area, rather than in-migrants, and are concentrated among young males located within 20 km of the mines. The two responses operate on different individuals through different channels: property declines reflect the rising opportunity cost of crime among those with weak baseline labor market attachment, while substance increases reflect an income effect among individuals already involved in substance use and directly employed in mining. I find no evidence that changes in the population composition due to migration, income inequality, occupational stress, and the government's crime-prevention capacity (police force) drive the crime results.

The findings carry several implications for policy in resource-dependent regions. First, the evidence supports labor market access through improved employment and earnings as the primary lever for reducing property crime among existing residents, without any targeted criminal justice intervention. This implies that place-based employment programs, active labor market policies, and industrial development that reaches the young male residential population can serve simultaneously as crime-prevention tools. Resources devoted to property crime enforcement may be partially displaced as labor market conditions improve, and could instead be reallocated toward the areas where the boom generates new social costs. Second, the flip side of this finding is that positive economic shocks create new demands on narcotics enforcement and public health infrastructure. Rather than expanding criminal justice responses to substance offenses, the evidence is more consistent with public health interventions in the sense that substance use among employed, high-income young males in a high-pressure work environment may be better addressed through occupational health programs, addiction prevention, and accessible treatment services than through increased policing. The detection channel analysis confirms that LKAB's internal testing policy already routes positive drug test results to rehabilitation rather than criminal referral.

Although the analysis focuses on a specific mining boom in northern Sweden, a developed-country mining context, the mechanisms studied in this paper are not unique to this context. The theoretical channels linking economic opportunities to criminal behavior, through changes in labor market returns, opportunity costs, disposable income, and population mobility, are central to a wide range of local economic shocks. Mining booms provide a particularly clean setting to study these mechanisms because they generate large, localized, and plausibly exogenous changes in economic conditions. However, similar dynamics arise in other settings characterized by rapid local economic expansion, including energy booms, infrastructure investments, manufacturing expansions, and large place-based development programs. At the same time, institutional features such as labor market regulation, social insurance, and law enforcement capacity may shape how these mechanisms operate, implying that the magnitude of the effects may vary across contexts even when the underlying behavioral responses are similar. Future work could examine how these mechanisms operate in settings with different institutional environments, such as developing countries or regions with weaker labor market protections and enforcement capacity.

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Online Appendix

Digging for Trouble?
Mining and Criminal Behavior of Young Males

Gabriel Rodríguez-Puello[1]

[1] Jönköping International Business School (JIBS), Jönköping University

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A Appendix: Additional tables

Table A.1: Mining municipalities, mines and mining employment share

County	Municipality	Mine(s) and main product(s)	Population 2015	Mining employment share 2003	2010	2015
Norrbotten	Gällivare	Malmberget (Iron ore) and Aitik (Copper)	18,123	17.44%	20.89%	22.56%
Norrbotten	Kiruna	Kirunavaara (Iron ore) and Gruvberget (Iron ore)	23,178	13.94%	16.51%	18.44%
Västerbotten	Lycksele	Kristineberg (Copper/zinc) and Svartliden (Gold)	12,177	1.50%	1.97%	1.70%
Västerbotten	Malå	Storliden (Zinc/copper)	3,109	4.93%	6.10%	7.64%
Västerbotten	Norsjö	Maurliden (Copper/zinc) and Maurliden Ö (Copper/zinc)	4,176	2.92%	2.68%	4.69%
Västerbotten	Skelleftea	Björkdal (Gold) and Renström (Copper/zinc)	72,031	1.82%	1.88%	2.61%
Västerbotten	Sorsele	Blaiken (Zinc)	2,516	0.52%	1.37%	0.99%
Västerbotten	Storuman	Svartliden (Gold) and Blaiken (Zinc)	5,943	0.75%	0.80%	1.07%
Örebro	Askersund	Zinkgruvan (Zinc)	11,151	7.24%	7.39%	7.75%
Dalarna	Hedemora	Garpenberg (Zinc)	15,235	3.24%	3.35%	4.40%

Notes: Information from Statistics Sweden, Nordregio (2009), SGU (2014), Tano et al. (2016), and SGU (2021). Following Tano et al. (2016), municipalities are considered if they had an operating mine during the mining boom ranging from 2004 to 2010. Only individuals located in Norrbotten County are included in the paper, either as treated or control. Employment in the mining sector via the Swedish Standard Industrial (SNI) Classification 2002 includes the codes 10100, 10200, 10301, 10302, 12000, 13100, 13200, 14110, 14120, 14130, 14210, 14220, 14300, 14400, 14500, 29520, and 51820.

Table A.2: Description of crime variables

Crime category	Description
Total violations of the criminal code	Violent crimes Property crimes Crimes against the public Crimes against the state
Violent crimes	1+2+3+4+5
1. Violations of life and health	Completed murder, manslaughter, or assault with fatal outcome. Attempt, preparation, and branding for murder or manslaughter. Child killing.
2. Violations of freedom and peace	Kidnapping, human trafficking, human exploitation. Illegal restraint. Child welfare violation. Unlawful coercion. Serious breach of peace, serious breach of women's peace, unlawful persecution. Unlawful threats. Unlawful use of identity. Illegal invasion of privacy. Molestation. Urge to commit suicide. Reckless solicitation of suicide. Data breach, illegal wiretapping.
3. Defamation	Crime of defamation. Slander, insult, slander of the deceased.
4. Sexual offenses	Rape incl. Bearish. Negligent rape. Rape against children incl. Bearish. Sexual assault incl. gross, negligent sexual assault. Sexual exploitation of children under the age of 18. Sexual assault incl. violently against children under the age of 18. Intercourse with offspring or siblings. Purchase of sexual services, pimping of persons 18 years and older. Exploitation of children for sexual posing, purchase of sexual act of children under 18 years. Sexual harassment, exhibitionism. Contact to meet a child for sexual purposes.
5. Crimes against family	Bigamy, illicit marriage; Undue influence in the adoption of children adoption of children; Distortion of family status.
Property crimes	1+2+3+4+5
1. Theft, robbery and other assault	Theft of motor-driven means of transport. Theft of non-motorized means of transportation. Theft (including burglary). Theft by burglary. Theft without breaking and entering. Robbery without a firearm. Robbery with a firearm.

	Other offenses against the Criminal Code.
2. Fraud and other misconduct	Fraud, petty fraud, gross fraud, gross debt fraud. Other offenses against the Criminal Code.
3. Embezzlement and other faithlessness	Embezzlement, petty embezzlement, gross embezzlement; Wrongful disposal; Misdemeanor; Breach of trust; Abuse of authority.
4. Offenses against creditors, etc.	Misconduct against creditors, gross misconduct against creditors; Aggravation of bankruptcy and executive proceedings; Negligence against creditors; Undue favoring of creditor.
5. Crime of damage	Damage, minor damage, injury, serious damage: on motor vehicles, car fire or other motor vehicle fire, by fire, against state, municipality, county council, other manage, graffiti against public transport.

Notes: Own elaboration using Brå (2023) as a base. For a detailed description of the types of crimes and the Swedish criminal code, consult Brå (2023).

Table A.3: Summary statistics, 2000-2003 and 2004-2015

	Control 2000-2003 Mean SD	Treated 2000-2003 Mean SD	Total 2000-2003 Mean SD	Control 2004-2015 Mean SD	Treated 2004-2015 Mean SD	Total 2004-2015 Mean SD
Convicted property crime	0.009	0.009	0.009	0.011	0.009	0.010
	(0.092)	(0.097)	(0.093)	(0.102)	(0.094)	(0.101)
Convicted violent crime	0.005	0.005	0.005	0.006	0.006	0.006
	(0.069)	(0.068)	(0.069)	(0.076)	(0.076)	(0.076)
Convicted substance crimes	0.002	0.002	0.002	0.007	0.007	0.007
	(0.047)	(0.042)	(0.047)	(0.086)	(0.084)	(0.086)
Convicted traffic crimes	0.008	0.010	0.008	0.009	0.010	0.009
	(0.088)	(0.098)	(0.090)	(0.094)	(0.097)	(0.094)
Married	0.184	0.150	0.178	0.158	0.125	0.153
	(0.387)	(0.357)	(0.383)	(0.365)	(0.331)	(0.360)
Children under 18	0.416	0.411	0.416	0.390	0.375	0.387
	(0.493)	(0.492)	(0.493)	(0.488)	(0.484)	(0.487)
Primary education	0.493	0.602	0.511	0.341	0.390	0.349
	(0.500)	(0.490)	(0.500)	(0.474)	(0.488)	(0.477)
Secondary education	0.409	0.348	0.398	0.551	0.553	0.551
	(0.492)	(0.476)	(0.490)	(0.497)	(0.497)	(0.497)
Tertiary education	0.099	0.050	0.090	0.108	0.057	0.100
	(0.298)	(0.218)	(0.287)	(0.311)	(0.232)	(0.299)
Non-employed	0.270	0.232	0.263	0.251	0.162	0.236
	(0.444)	(0.422)	(0.440)	(0.433)	(0.369)	(0.425)
Primary sector	0.024	0.205	0.054	0.028	0.244	0.064
	(0.152)	(0.404)	(0.226)	(0.164)	(0.429)	(0.244)
Secondary sector	0.268	0.190	0.255	0.269	0.232	0.263
	(0.443)	(0.392)	(0.436)	(0.444)	(0.422)	(0.440)
Tertiary sector	0.439	0.373	0.428	0.452	0.362	0.437
	(0.496)	(0.484)	(0.495)	(0.498)	(0.481)	(0.496)
Nxt	100410	20184	120594	266136	53672	319808
N	25102	5046	30148	22178	4473	26651

Notes: The table shows means and standard deviations for the treated (Gällivare/Kiruna) and control (other Norrbotten County) groups separately for the pre-boom (2000–2003) and post-boom (2004–2015) periods. Economic sector follows SNI 2002: primary = extraction/agriculture; secondary = manufacturing/construction; tertiary = services, healthcare, public sector, and other. Marital status is a dummy variable equal to 1 if married. Education is categorized as primary, secondary, and tertiary. N decreases from pre- to post-boom as individuals age out of the 18–29 window. Sample: young male residents aged 18–29, migrants excluded. Data from administrative registers, Statistics Sweden (LISA).

Table A.4: Mean differences of changes (2000-2003) comparing treated and control individuals

	Treated	Control
Convicted property crime	0.00	0.00
Convicted violent crime	0.00	0.00
Convicted substance crime	0.00	0.00
Convicted traffic crime	0.00	0.00
Married	0.03	0.05^{***}
Children under 18	0.03	0.03
Primary education	-0.06	-0.06
Secondary education	0.05	0.01^{***}
Tertiary education	0.02	0.05^{***}
Non-employed	-0.07	-0.06
Primary sector	0.03	-0.00^{***}
Secondary sector	-0.01	0.02^{***}
Tertiary sector	0.05	0.05

Notes: The table shows the change in each individual characteristic between 2000 and 2003 for treated and control individuals, used to assess pre-boom parallel trends in observable characteristics. Stars indicate the statistical significance of the difference in these pre-boom changes between groups. No economically meaningful differences are found for the crime outcomes, supporting the parallel trends assumption. Marital status is a dummy variable equal to 1 if married. Education is categorized as primary, secondary, and tertiary. The economic sectors are divided into primary (extraction and agricultural sector), secondary (manufacturing and construction), and tertiary (services, healthcare, public sector, and others). Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Impact of the mining boom on property crimes by offense type, 2000-2015

	(1) Theft crimes	(2) Fraud crimes	(3) Embezzlement crime	(4) Crimes to creditors	(5) Vandalism crime
Post*Treated	-0.0023	-0.0007	-0.0027**	-0.0000	-0.0025*
	(0.0019)	(0.0009)	(0.0011)	(0.0006)	(0.0015)
Individual FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480	230480
N	14405	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0067	0.0024	0.0016	0.0008	0.0031
Effect relative to the mean (%)	-33.93	-29.66	-170.07	-2.78	-78.45
R-squared	0.2924	0.2122	0.1685	0.1561	0.1945
Within R-squared	0.0000	0.0000	0.0001	0.0000	0.0000

Notes: The table disaggregates the property crime result from Table 1 into five legal subcategories: theft and robbery (Ch. 8), fraud (Ch. 9), embezzlement and breach of trust (Ch. 10), offenses against creditors (Ch. 11), and property damage/vandalism (Ch. 12). Each outcome is a binary conviction indicator. Design and sample identical to Table 1 (14,405 individuals, 230,480 person-years). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Impact of the mining boom on narcotics-related suspicions by offense type, 2000-2015

	(1) Narcotic suspicion	(2) Narcotic production	(3) Narcotic distribution	(4) Narcotic use-holding
Post*Treated	0.0055**	0.0009**	0.0014*	0.0049**
	(0.0023)	(0.0004)	(0.0007)	(0.0023)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480
N	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0045	0.0000	0.0006	0.0017
Effect relative to the mean (%)	122.92	2603.78	209.61	290.00
R-squared	0.4056	0.1691	0.2194	0.4025
Within R-squared	0.0000	0.0000	0.0000	0.0000

Notes: The table disaggregates narcotics-related criminal activity using data from the Crime Suspicion Register (BRÅ) rather than the conviction records used in the main analysis. Suspicion data capture police-recorded incidents regardless of prosecution outcome and allow decomposition into: any narcotics suspicion (Col. 1), production (Col. 2), distribution/selling (Col. 3), and holding/use (Col. 4). Base rates for production and distribution are very low; estimates should be interpreted with caution. Design and sample identical to Table 1. Standard errors (in parentheses) clustered at the grid level. Data from the BRÅ Crime Suspicion Register and administrative registers, Statistics Sweden. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Impact of the mining boom on substance crime by pre-boom property crime history, 2000-2015

	(1) Had property crime pre	(2) No property crime pre
Post*Treated	0.0230	0.0036**
	(0.0252)	(0.0015)
Individual FE	Yes	Yes
Year FE	Yes	Yes
Grid FE	Yes	Yes
Nxt	5080	225400
N	318	14088
Mean dep. var (2000-03)	0.0471	0.0008
Effect relative to the mean (%)	48.78	432.42
R-squared	0.4931	0.3667
Within R-squared	0.0003	0.0000

Notes: The table tests whether the increase in substance crime reflects crime switching by the same individuals who reduced property crime. The sample is split by pre-boom property crime history: Column (1) includes individuals with at least one property crime conviction in 2000–2003 (N = 318); Column (2) includes those with no pre-boom property crime convictions (N = 14,088). If switching drives the results, the substance crime increase should be concentrated in Column (1). The small sample in Column (1) limits statistical power. Design otherwise identical to Table 1. Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Impact of the mining boom on criminal behavior by mining sector transition, 2000-2015

	(1) Property crime (entrants)	(2) Property crime (non-entrants)	(3) Substance crime (entrants)	(4) Substance crime (non-entrants)
Post*Treated	-0.0107**	-0.0045	0.0000	0.0057**
	(0.0048)	(0.0027)	(0.0019)	(0.0022)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	196544	225951	196544	225951
N	12284	14122	12284	14122
Mean dep. var (2000-03)	0.0121	0.0124	0.0024	0.0026
Effect relative to the mean (%)	-88.47	-36.06	0.95	216.20
R-squared	0.2959	0.2947	0.3782	0.3769
Within R-squared	0.0000	0.0000	0.0000	0.0001

Notes: The table splits treated individuals by whether they newly entered the mining sector after the boom. Entrants (Cols 1, 3) were not in mining in 2003 but worked in mining in at least one year during 2004–2015. Non-entrants (Cols 2, 4) did not newly enter mining post-2004, a group that includes those already in mining before the boom. The full control group is included in all columns. Because sector entry after 2004 is endogenous to the boom, these estimates are descriptive and should not be interpreted causally. Design otherwise identical to Table 1. Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: Impact of the mining boom on criminal behavior by distance to the mines, 2000-2015

	(1) Property crime	(2) Violent crime	(3) Traffic crime	(4) Substance crime
Post*≤ 20 km	-0.0076***	0.0015	-0.0018	0.0055***
	(0.0029)	(0.0017)	(0.0021)	(0.0020)
Post* 20 - 40 km	-0.0019	-0.0002	-0.0046	-0.0072*
	(0.0057)	(0.0036)	(0.0116)	(0.0042)
Post*40 - 60 km	0.0078	0.0090	-0.0024	0.0000
	(0.0089)	(0.0060)	(0.0125)	(0.0015)
Post*60 - 80 km	0.0007	0.0027	-0.0046	-0.0008
	(0.0051)	(0.0039)	(0.0078)	(0.0017)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480
N	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0125	0.0060	0.0097	0.0025
R-squared	0.2929	0.2195	0.2708	0.3759
Within R-squared	0.0001	0.0000	0.0000	0.0001

Notes: The table reports the coefficient estimates underlying Figure A.9. Treatment is defined by 20-kilometer distance rings from the nearest mine; residents beyond 80 km are the reference category. Crime definitions are in Table 1. Design and sample identical to Table 1 (14,405 individuals, 230,480 person-years). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Summary statistics of residents and migrants, 2000-2003 and 2004-2015

	Residents 2000-2003	Residents 2004-2015	Migrants (county) 2000-2003	Migrants (county) post-migration-2015	Migrants (Treated mun.) 2000-2003	Migrants (Treated mun.) post-migration-2015
	Mean/SD	Mean/SD	Mean/SD	Mean/SD	Mean/SD	Mean/SD
Convicted of property crime	0.01	0.01	0.07	0.04	0.01	0.01
	(0.10)	(0.09)	(0.25)	(0.19)	(0.11)	(0.11)
Convicted of violent crime	0.00	0.01	0.02	0.01	0.00	0.00
	(0.07)	(0.08)	(0.15)	(0.11)	(0.06)	(0.07)
Convicted of traffic crime	0.01	0.01	0.05	0.03	0.02	0.01
	(0.10)	(0.10)	(0.22)	(0.16)	(0.13)	(0.09)
Convicted of substance crime	0.00	0.01	0.05	0.04	0.00	0.01
	(0.04)	(0.08)	(0.22)	(0.19)	(0.05)	(0.10)
Disposable income(100SEK)	1439.84	1886.59	1080.36	1372.35	1015.39	1907.15
	(792.69)	(1244.01)	(3148.50)	(1321.92)	(1096.85)	(1059.97)
Yearly earnings(100SEK)	1701.11	2278.91	978.01	1367.14	1000.87	2281.41
	(1172.99)	(1453.96)	(1317.49)	(1521.95)	(954.14)	(1487.67)
Employment	0.77	0.84	0.51	0.60	0.59	0.82
	(0.42)	(0.37)	(0.50)	(0.49)	(0.49)	(0.38)
Age	30.13	28.26	27.31	27.57	24.85	29.03
	(6.44)	(6.65)	(6.20)	(5.53)	(5.78)	(5.44)
Married	0.15	0.13	0.11	0.12	0.06	0.14
	(0.36)	(0.33)	(0.31)	(0.33)	(0.23)	(0.34)
Children under 18	0.41	0.37	0.24	0.23	0.30	0.28
	(0.49)	(0.48)	(0.43)	(0.42)	(0.46)	(0.45)
Primary education	0.60	0.39	0.47	0.26	0.34	0.22
	(0.49)	(0.49)	(0.50)	(0.44)	(0.47)	(0.41)
Secondary education	0.35	0.55	0.44	0.52	0.60	0.53
	(0.48)	(0.50)	(0.50)	(0.50)	(0.49)	(0.50)
Tertiary education	0.05	0.06	0.10	0.22	0.06	0.25
	(0.22)	(0.23)	(0.29)	(0.42)	(0.23)	(0.43)
Non-employed	0.23	0.16	0.49	0.40	0.41	0.18

	(0.42)	(0.37)	(0.50)	(0.49)	(0.49)	(0.38)
Primary economic sector	0.21	0.24	0.01	0.02	0.04	0.17
	(0.40)	(0.43)	(0.10)	(0.13)	(0.21)	(0.38)
Secondary economic sector	0.19	0.23	0.14	0.13	0.12	0.17
	(0.39)	(0.42)	(0.35)	(0.34)	(0.32)	(0.38)
Tertiary economic sector	0.37	0.36	0.36	0.45	0.43	0.48
	(0.48)	(0.48)	(0.48)	(0.50)	(0.49)	(0.50)

Notes: The table compares demographic and labor market characteristics across three groups: residents in Gällivare/Kiruna (Cols 1–2), all migrants to Norrbotten County in 2004 or later (Cols 3–4), and migrants to the mining municipalities specifically (Cols 5–6). For residents, pre/post refers to 2000–2003 and 2004–2015; for migrants, pre/post refers to before and after the move. Unlike the main analysis, this table uses the full 18–39-year-old male sample to provide a broader picture of migration patterns. Migrants are assumed to have moved in response to the mining boom's improved labor market conditions. Marital status is a dummy variable equal to 1 if married. Education is categorized as primary, secondary, and tertiary. The economic sectors are divided into primary (extraction and agricultural sector), secondary (manufacturing and construction), and tertiary (services, healthcare, public sector, and others). Data from administrative registers, Statistics Sweden (LISA).

Table A.11: Impact of the mining boom on criminal behavior by rings, 2000-2015

	(1) Property crime	(2) Violent crime	(3) Traffic crime	(4) Substance crime
Post*Ring 1	-0.0105	0.0009	-0.0062	0.0091*
	(0.0086)	(0.0032)	(0.0042)	(0.0049)
Post*Ring 2	-0.0106**	-0.0019	-0.0046	0.0041
	(0.0045)	(0.0049)	(0.0055)	(0.0046)
Post*Ring 3	0.0037	0.0068*	0.0019	0.0061***
	(0.0042)	(0.0039)	(0.0039)	(0.0021)
Post*Ring 4	-0.0097	0.0037	0.0017	0.0050
	(0.0060)	(0.0042)	(0.0033)	(0.0041)
Post*Ring 5	-0.0070	-0.0045	-0.0055	0.0053
	(0.0046)	(0.0033)	(0.0051)	(0.0058)
Post*Ring 6	-0.0148**	0.0038	0.0034	0.0018
	(0.0069)	(0.0025)	(0.0045)	(0.0033)
Post*Ring 7	0.0019	0.0054*	-0.0028	-0.0032**
	(0.0040)	(0.0029)	(0.0074)	(0.0016)
Post*Ring 8	-0.0029	0.0015	-0.0004	0.0009
	(0.0058)	(0.0035)	(0.0055)	(0.0021)
Post*Ring 9	-0.0010	-0.0042	-0.0055	-0.0004
	(0.0057)	(0.0041)	(0.0063)	(0.0017)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480
N	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0125	0.0060	0.0097	0.0025
R-squared	0.2929	0.2195	0.2708	0.3759
Within R-squared	0.0001	0.0001	0.0000	0.0001

Notes: The table replicates the distance analysis using ten granular rings based on the actual distance distribution rather than fixed 20-kilometer intervals. Ring boundaries (km): Ring 1: 0.00–2.74; Ring 2: 2.75–3.37; Ring 3: 3.38–3.82; Ring 4: 3.83–4.26; Ring 5: 4.27–4.86; Ring 6: 4.87–18.34; Ring 7: 18.35–73.67; Ring 8: 73.68–102.47; Ring 9: 102.48–125.37; Ring 10: 125.38–236.00 km (reference). Design and sample identical to Table 1. Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.12: Impact of the mining boom on criminal behavior using time duration for treatment, 2000-2015

	(1) Property crime	(2) Violent crime	(3) Traffic crime	(4) Substance crime
Post*$\leq$ 20 min	-0.0128***	0.0038	-0.0048	0.0083**
	(0.0046)	(0.0026)	(0.0031)	(0.0035)
Post* 20 - 40 min	0.0056	-0.0008	-0.0025	-0.0061
	(0.0070)	(0.0042)	(0.0120)	(0.0048)
Post*40 - 60 min	-0.0027	-0.0002	-0.0003	0.0029
	(0.0029)	(0.0021)	(0.0026)	(0.0021)
Post*60 - 80 min	0.0025	0.0076	-0.0130	-0.0053***
	(0.0097)	(0.0086)	(0.0166)	(0.0017)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	230480	230480	230480	230480
N	14405	14405	14405	14405
Mean dep. var (2000-03)	0.0125	0.0060	0.0097	0.0025
R-squared	0.2929	0.2195	0.2708	0.3759
Within R-squared	0.0001	0.0000	0.0000	0.0001

Notes: The table replicates Table A.9 using driving time by car to the nearest mine (in minutes) as the spatial treatment measure, rather than straight-line distance in kilometers. Rings are defined in 20-minute intervals: ≤20, 20–40, 40–60, and 60–80 minutes, with those beyond 80 minutes as the reference. Design and sample identical to Table 1 (14,405 individuals, 230,480 person-years). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.13: Robustness checks: impact of the mining boom on criminal behavior, 2000-2015

	(1) Baseline Residents	(2) Residents (treated 2003)	(3) Including controls	(4) Residents and migrants	(5) Balanced panel	(6) Exclude neigh. municipalities	(7) Municipality fixed-effect
Panel A: Property crime							
Post*Treated	-0.0066**		-0.0060**	-0.0052**	-0.0061*	-0.0074***	-0.0061**
	(0.0027)		(0.0027)	(0.0024)	(0.0037)	(0.0027)	(0.0025)
Post*Treated (2003)		-0.0045*					
		(0.0027)					
Mean dep. var (2003)	0.0125	0.0125	0.0125	0.0127	0.0139	0.0123	0.0125
Effect relative to the mean (%)	-52.40	-35.86	-48.08	-41.41	-44.04	-60.26	-48.45
R-squared	0.2929	0.2991	0.2932	0.3020	0.2800	0.2951	0.2710
Within R-squared	0.0001	0.0000	0.0005	0.0000	0.0001	0.0001	0.0000
Panel B: Violent crime							
Post*Treated	0.0018		0.0019	0.0014	0.0026	0.0019	0.0014
	(0.0016)		(0.0016)	(0.0015)	(0.0019)	(0.0016)	(0.0016)
Post*Treated (2003)		0.0019					
		(0.0017)					
Mean dep. var (2003)	0.0060	0.0060	0.0060	0.0061	0.0066	0.0059	0.0060
Effect relative to the mean (%)	29.18	31.34	31.30	23.53	39.85	31.77	22.53
R-squared	0.2195	0.2084	0.2196	0.2213	0.2104	0.2219	0.1975
Within R-squared	0.0000	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000
Panel C: Traffic crime							
Post*Treated	-0.0017		-0.0020	-0.0026	-0.0037	-0.0024	-0.0016
	(0.0020)		(0.0020)	(0.0018)	(0.0030)	(0.0020)	(0.0019)
Post*Treated (2003)		-0.0010					
		(0.0020)					
Mean dep. var (2003)	0.0097	0.0097	0.0097	0.0097	0.0112	0.0089	0.0097
Effect relative to the mean (%)	-17.06	-10.45	-20.16	-26.61	-32.71	-27.19	-16.16
R-squared	0.2708	0.2815	0.2711	0.2735	0.2697	0.2744	0.2480
Within R-squared	0.0000	0.0000	0.0004	0.0000	0.0000	0.0000	0.0000

			Panel D: Substance crime				
Post*Treated	0.0046**		0.0048***	0.0042**	0.0019	0.0047**	0.0039**
	(0.0018)		(0.0018)	(0.0017)	(0.0020)	(0.0018)	(0.0017)
Post*Treated (2003)		0.0046**					
		(0.0018)					
Mean dep. var (2003)	0.0025	0.0025	0.0025	0.0027	0.0028	0.0023	0.0025
Effect relative to the mean (%)	181.24	179.33	188.73	154.33	68.23	202.28	154.28
R-squared	0.3759	0.3560	0.3764	0.3769	0.3662	0.3765	0.3600
Within R-squared	0.0000	0.0001	0.0009	0.0000	0.0000	0.0001	0.0000
Nxt	230480	124267	230480	263626	49698	193151	230480
N	14405	7767	14405	16477	3106	12072	14405
Controls	No	No	Yes	No	No	No	No
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Municipality FE	No	No	No	No	No	No	Yes

Notes: Each column presents a robustness variant of the main result in Table 1, organized in four panels by crime type. Col. (1): Baseline (= Table 1). Col. (2): Treatment status assigned using the 2003 municipality. Col. (3): Adds time-varying individual controls (marital status, children under 18, education, economic sector); controls are excluded from the main specification because some may be endogenous to the boom. Col. (4): Includes migrants alongside residents (16,477 individuals). Col. (5): Balanced panel — individuals observed in all 16 years (3,106 individuals). Col. (6): Excludes the four Norrbotten municipalities nearest to treated areas (Jokkmokk, Pajala, Överkalix, Boden) to address spillover concerns. Col. (7): Replaces grid fixed effects with municipality fixed effects. Standard errors (in parentheses) clustered at the grid level. Crime definitions as in Table 1. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.14: Mechanisms: impact of the mining boom on wages and work intensity, 2000-2015

	(1) Monthly wage	(2) Log monthly wage	(3) Work scope
Post*Treated	933.0368***	0.0342***	-2.3489***
	(153.2177)	(0.0074)	(0.9019)
Individual FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes
Nxt	52020	52020	52020
N	3251	3251	3251
Mean dep. var (2000-03)	18065.5247	9.7804	89.7336
Effect relative to the mean (%)	5.16	0.35	-2.62
R-squared	0.8184	0.8300	0.6649
Within R-squared	0.0015	0.0011	0.0003

Notes: The table reports DID estimates using employer-reported data from the Lönestrukturstatistik (LSS), Statistics Sweden's Structure of Earnings Survey. Monthly wage (Col. 1) and log monthly wage (Col. 2) are full-time equivalent monthly salary, capturing the wage rate independently of working-time arrangements. Work scope (Col. 3) is contracted working time as a percentage of full-time (0–100); a positive effect would indicate increased work intensity. Because the LSS targets enterprises with ten or more employees, the matched sample is smaller than the main analysis (3,251 individuals, 52,020 person-years). Design otherwise identical to Table 1. Standard errors (in parentheses) clustered at the grid level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

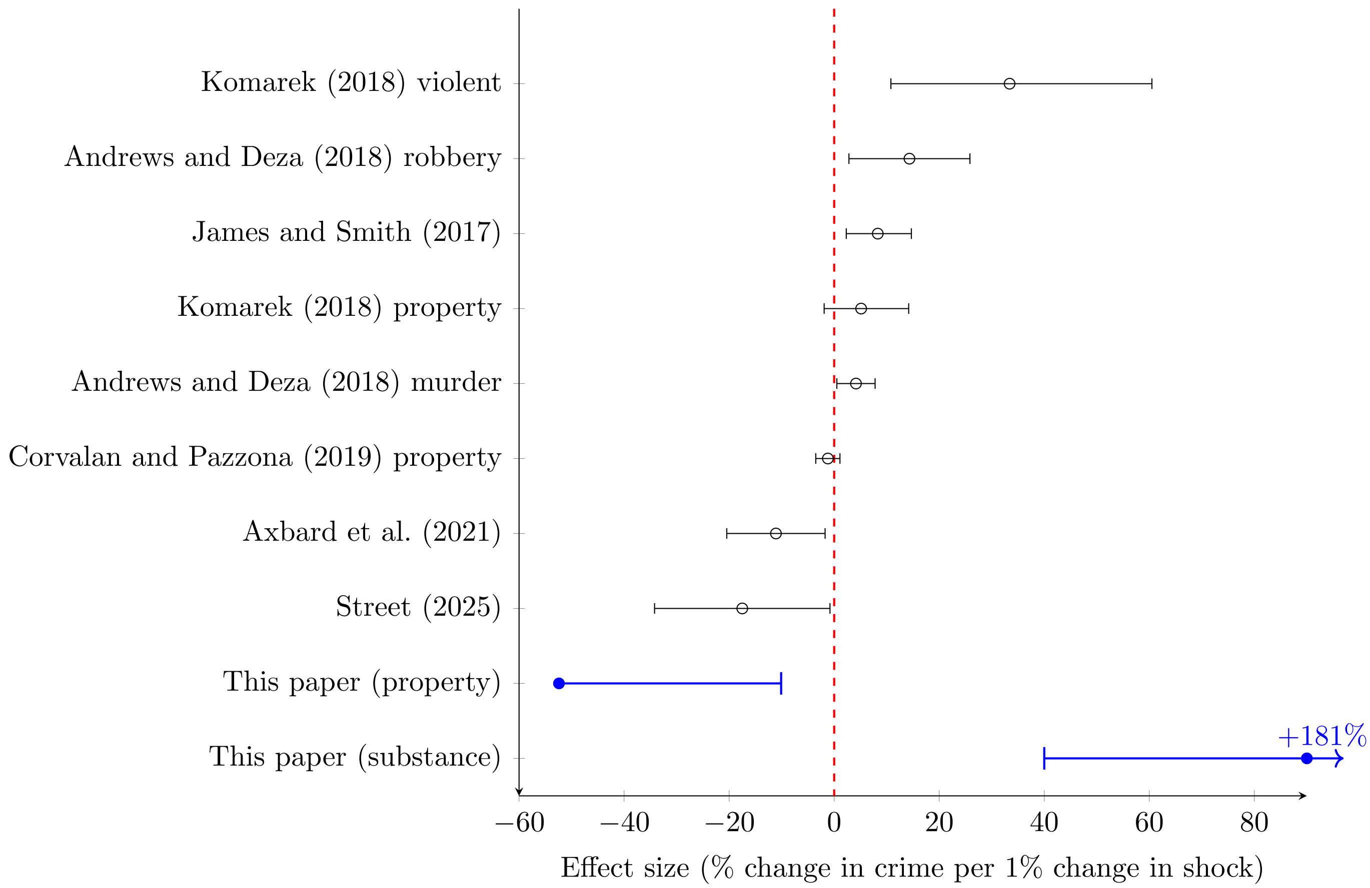
Table A.15: Social cost effects of the mining boom, 18-29-year-old male sample

	Property crime	Violent crime	Traffic crime	Substance crime
	Total social cost effect (1000s SEK)			
Post	-17866.61**	11345.70	-1921.11	14086.18**
	(7223.97)	(10332.34)	(2327.19)	(5473.34)

Notes: The table shows the social costs of effects computed using the DID estimates of the effect of the mining boom on different types of crime for young males. I take the total unit cost for each crime category for the UK using 2015/2016 prices and convert it to SEK using the 2004 exchange rate (1 GBP = 13.45 SEK). The total unit costs for property and violent crimes include the costs for anticipation (e.g., defensive expenditure), consequence (e.g., physical and emotional harm), and response (e.g., police costs) to the crime. The source for the crime costs is Heeks et al. (2018), which includes response costs (police, courts, probation) within each crime category. Incarceration costs are not reported as a separate row because the crime effects are concentrated among convictions without prison sentences. For property crimes, I use the estimated cost of 79,794 SEK for domestic burglary and dwelling. For violent crimes, I use the estimated cost of 189,057 SEK for violence with injury. For the traffic crimes, I use the estimated cost of 33,936 SEK, which accounts for a damage-only accident. For substance crimes, I use the estimated cost of 89,523 SEK, which only accounts for the cost of arrest. I multiply the estimated coefficients of Table 1 by the cost of crime and add it by the number of treated young male individuals during the boom, obtaining the aggregate social cost effect for this population (welfare implication). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

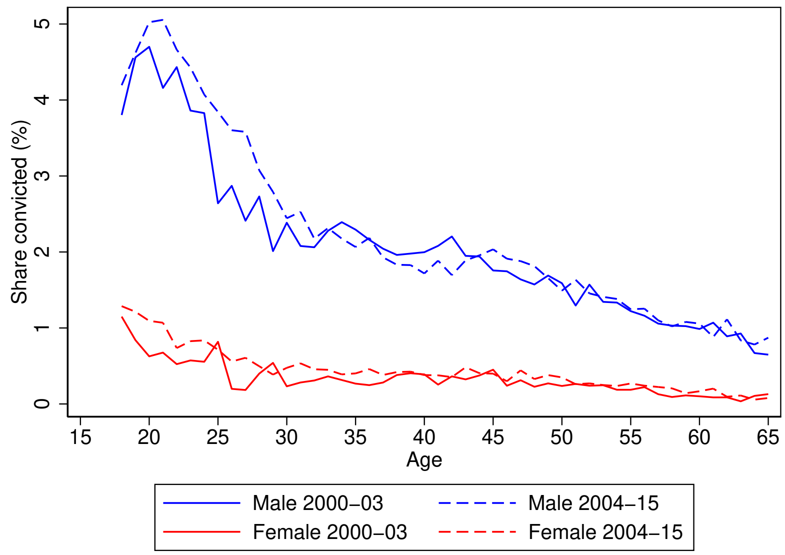
B Appendix: Additional figures

Figure B.1: Literature comparisons: resource shocks and crime



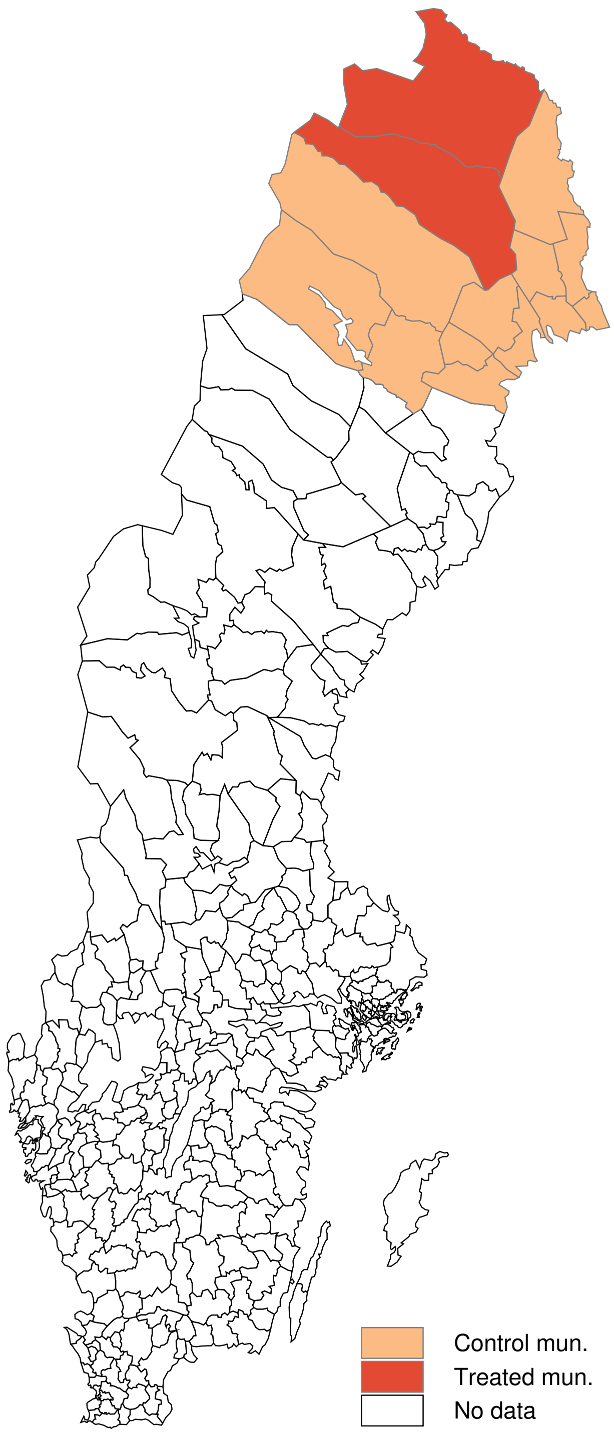
Notes: This figure compares estimated treatment effect sizes in the literature. Each dot shows the estimated effect size (%) with 95% confidence intervals. The figure compares the effect of different resource shocks exposure, such as mining and fracking booms, on criminal behavior. See Online Appendix E for details on the papers and effect size construction. Each point indicates the estimated effect of treatment (direct percent change) on criminal behavior for treated areas or individuals relative to controls as a percent of the control mean. When not specified, the outcome in the paper is all types of crime. This paper's estimates refer to young males aged 18–29; larger effect sizes relative to studies using broader samples are expected given this group's higher baseline crime rates and greater responsiveness to labor market conditions.

Figure B.2: Conviction rates of any crime by age and gender, before vs after



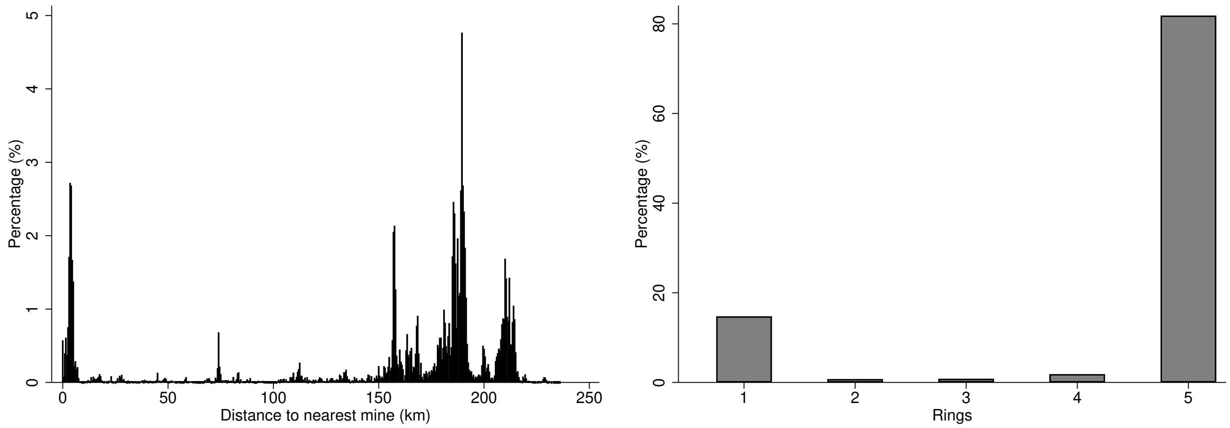
Notes: The figure plots the share of individuals convicted of any crime by single year of age (15–65) and gender, separately for the pre-boom (2000–2003) and post-boom (2004–2015) periods. The figure is descriptive and illustrates the age-crime profile motivating the focus on young males. Migrants are excluded. Data from administrative registers, Statistics Sweden (LISA).

Figure B.3: Treated and control municipalities



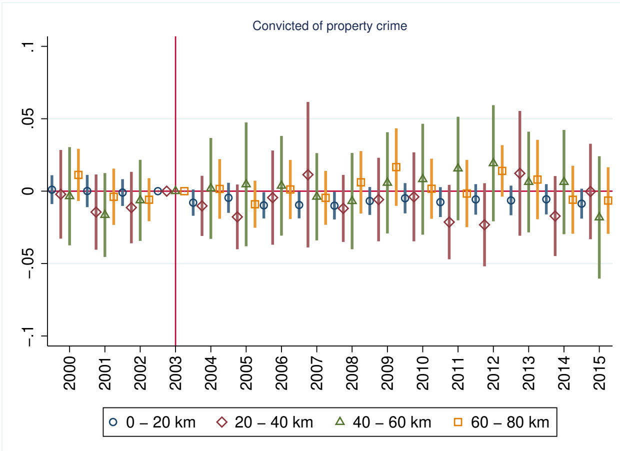
Notes: The map shows the spatial location of treated municipalities (Gällivare and Kiruna, darker shading) and control municipalities (other Norrbotten County municipalities, lighter shading). Municipalities in white are outside Norrbotten County and are excluded from the empirical sample. Data from Statistics Sweden.

Figure B.4: Distribution of individuals according to their distance to the nearest mine and in the rings

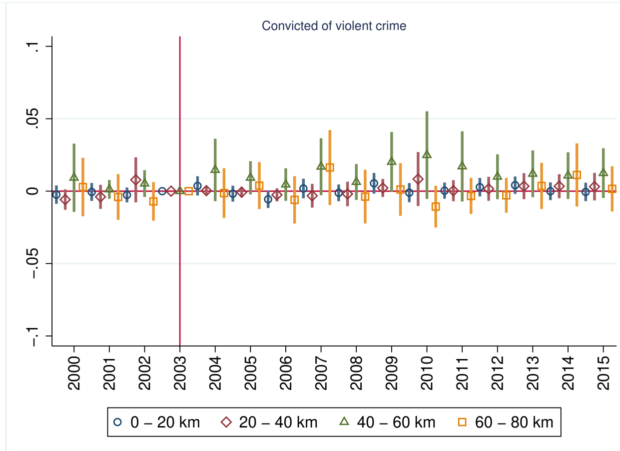


Notes: The left panel shows the distribution of sample individuals by distance (km) to the nearest of the three operating iron ore mines. The right panel shows the percentage in each of the five 20-kilometer rings used in Table A.9 and Figure 4. The majority of treated individuals live within 20 km of the nearest mine. Sample: young male residents aged 18–29, migrants excluded. Data from administrative registers and geocoded grid coordinates, Statistics Sweden.

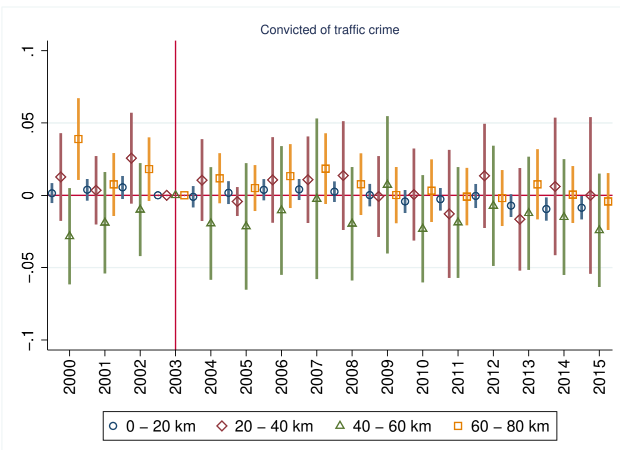
Figure B.5: Event study of the impact of the mining boom on criminal behavior by rings, 2000-2015



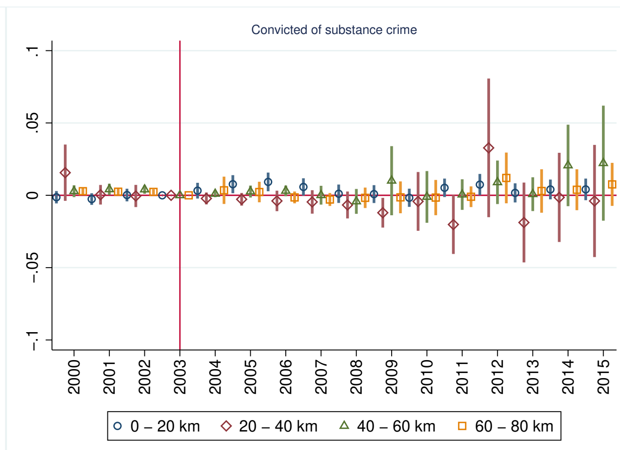
(a) Property crime



(b) Violent crime



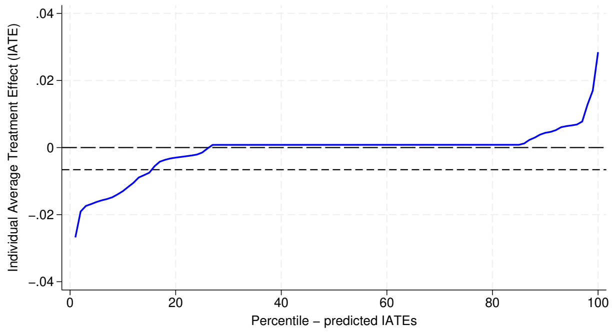
(c) Traffic crime



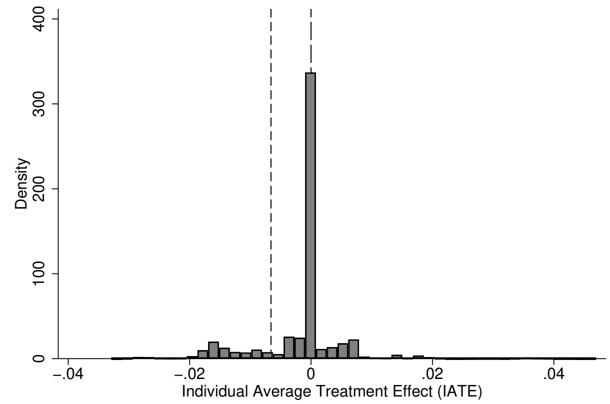
(d) Substance crime

Notes: The figure plots year-by-year DID coefficients from the dynamic model, estimated separately for each 20-kilometer distance ring. Each line traces the annual treatment effect for the indicated ring relative to residents beyond 80 km (reference) and the reference year 2003. 95% confidence interval shown. Pre-2004 coefficients test parallel trends within each ring. Design and sample identical to Table 1. Standard errors clustered at the grid level.

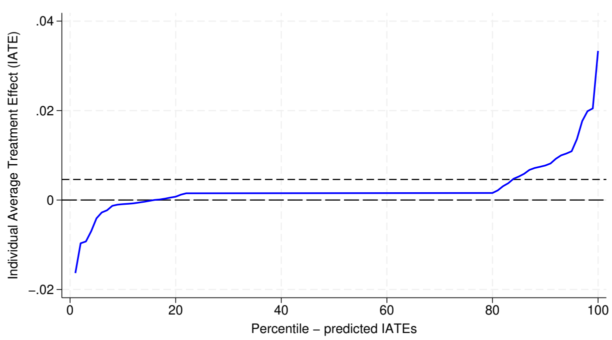
Figure B.6: Distribution predicted individual average treatment effects



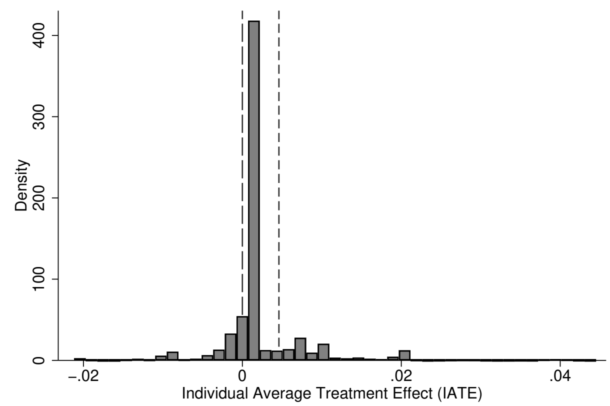
(a) Property crime



(b) Property crime



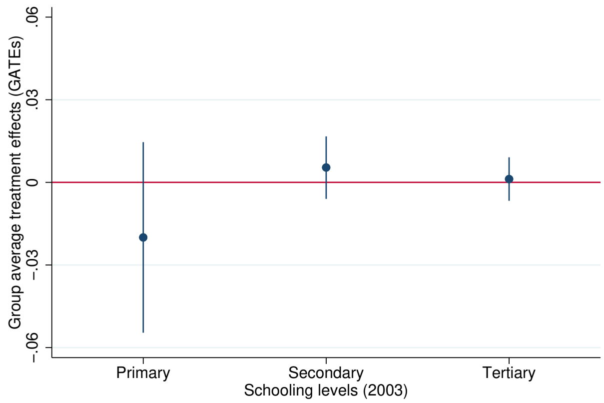
(c) Substance crime



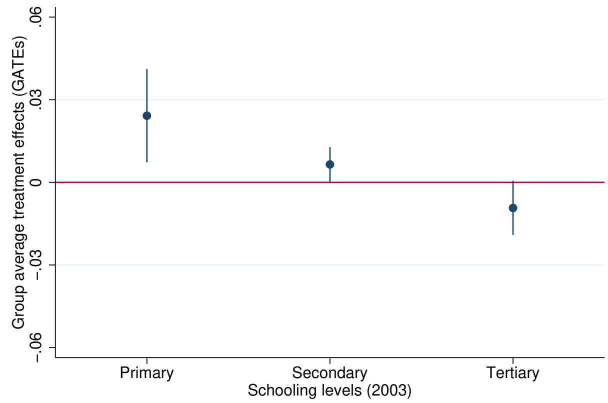
(d) Substance crime

Notes: The figure shows the distribution of Individual Average Treatment Effects (IATEs) from the causal forest, separately for property crime (panels a–b) and substance crime (panels c–d). Panels (a) and (c) plot IATEs ranked by percentile; panels (b) and (d) show the kernel density. The long-dashed line marks zero; the short-dashed line marks the causal forest ATE, which can be compared with the DID ATE from Table 1. Sample identical to Table 1; estimation details in Online Appendix D.

Figure B.7: Group average treatment effects (GATEs) by schooling levels, 2000-2015



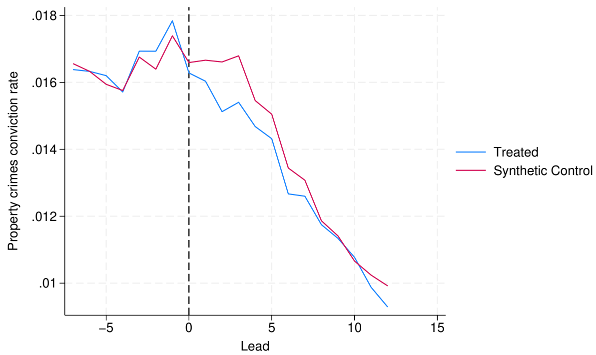
(a) Property crimes



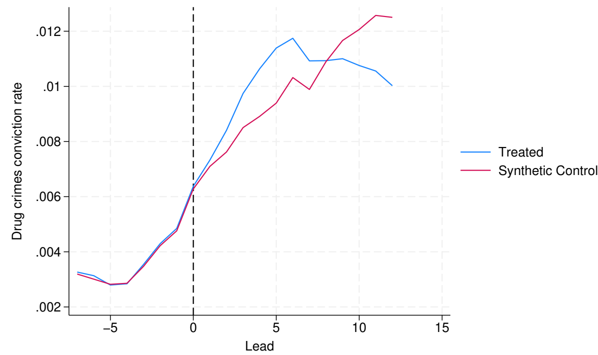
(b) Substance crimes

Notes: The figure replicates the schooling dimension of Figure 5a, restricting the sample to individuals aged 23–29 to separate low educational attainment from incomplete schooling among the youngest cohorts. GATEs for property crime (panel a) and substance crime (panel b) are shown by schooling level (primary/secondary/tertiary), measured in 2003. Bars show 95% confidence intervals. Estimation details in Online Appendix D.

Figure B.8: Synthetic control: Property and substance crime convictions



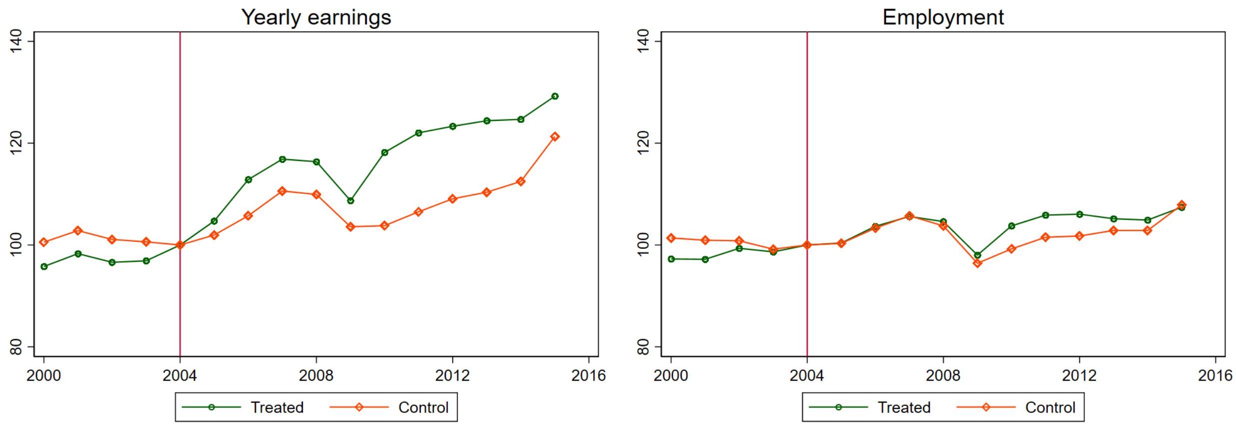
(a) Property crimes



(b) Substance crimes

Notes: The figure plots actual and synthetic control crime trajectories for the two treated mining municipalities (Gällivare and Kiruna), pooled into a single composite treated unit following Galiani and Quistorff (2017). The outcome is the 3-year centered moving average of the annual conviction rate, defined as the share of young males aged 18–29 convicted of the relevant crime type in a given municipality and year, collapsed from individual-level register data. The synthetic control is constructed as a weighted combination of control municipalities that minimizes the pre-treatment mean squared prediction error (MSPE) over 1996–2003, matching on all eight pre-treatment lags of the outcome variable. The donor pool excludes other mining municipalities, municipalities belonging to the Stockholm, Gothenburg, and Malmö metropolitan regions, and municipalities with steel industry employment shares above 10% in 2003. The horizontal axis shows years relative to treatment onset; the dashed vertical line marks the last pre-treatment year (2003).

Figure B.9: Earnings and employment evolution for mining and comparison municipalities, 2000-2015



(a) Yearly earnings (b) Employment

Notes: The figure plots average yearly earnings (panel a) and employment rates (panel b) for treated municipalities (Gällivare/Kiruna) and comparison municipalities (other Norrbotten County), both indexed to 100 in 2004. The vertical dashed line marks the start of the mining boom. The figure provides a visual overview of the first-stage labor market effects underlying Table 5. Sample: young male residents aged 18–29, migrants excluded. Data from administrative registers, Statistics Sweden (LISA).

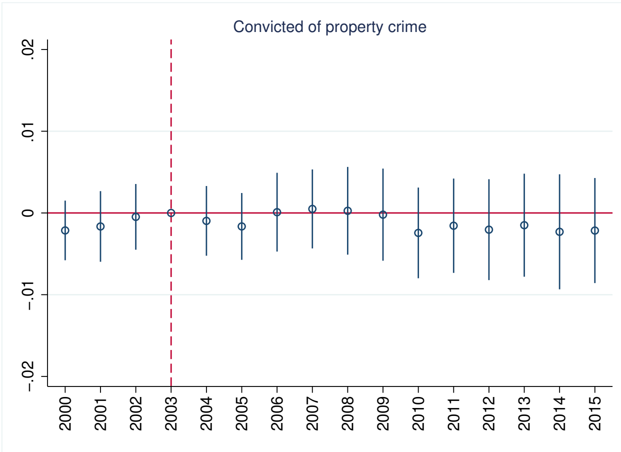
C Appendix: Results for the 30-39-year-old males sample

Table C.1: Impact of the mining boom on criminal behavior for 30-39 years old, 2000-2015

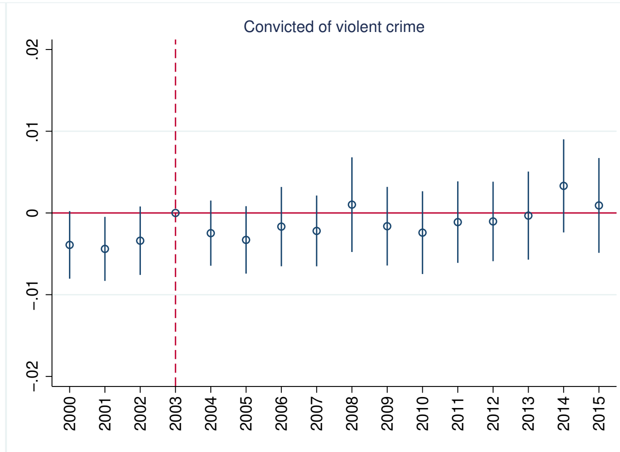
	(1) Property crime	(2) Violent crime	(3) Traffic crime	(4) Substance crime
Post*Treated	0.0002	0.0004	0.0008	-0.0008
	(0.0012)	(0.0011)	(0.0015)	(0.0006)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	209922	209922	209922	209922
N	13120	13120	13120	13120
Mean dep. var (2000-03)	0.0055	0.0036	0.0067	0.0019
Effect relative to the mean (%)	3.11	11.27	12.66	-42.73
R-squared	0.3088	0.2396	0.3071	0.4397
Within R-squared	0.0000	0.0000	0.0000	0.0000

Notes: The table replicates Table 1 for males aged 30–39 as a falsification exercise. Older males have stronger pre-existing labor market attachment and are expected to be less responsive to the boom than the main sample. The design is identical to Table 1 with the sample replaced by 30–39-year-old male residents (13,120 individuals, 209,922 person-years). Crime definitions as in Table 1. Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

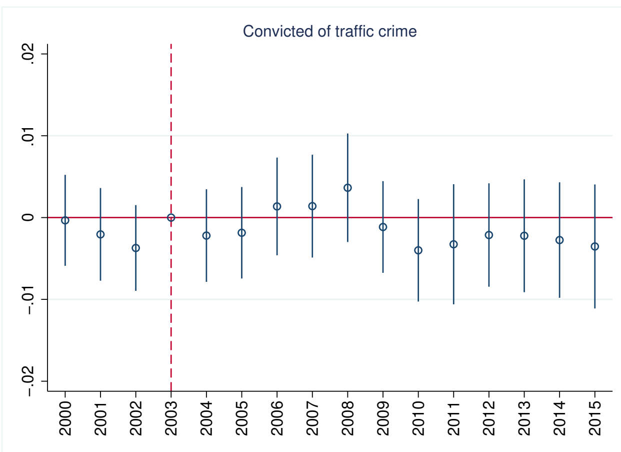
Figure C.1: Event study of the impact of the mining boom on criminal behavior of 30-39-year-old males, 2000-2015



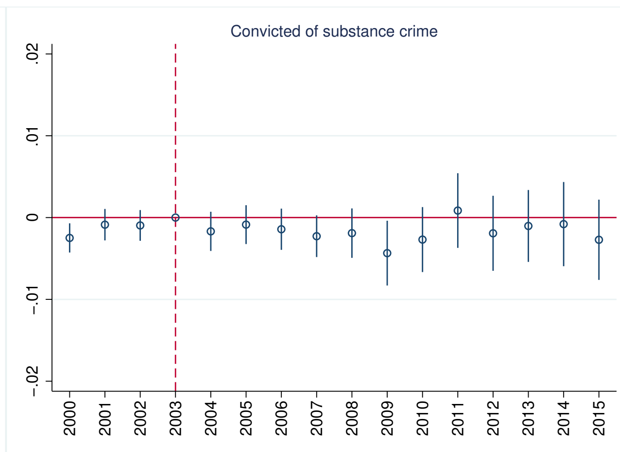
(a) Property crime



(b) Violent crime



(c) Traffic crime



(d) Substance crime

Notes: The figure replicates Figure 3 for males aged 30–39 as a falsification exercise. Year-by-year DID coefficients are plotted for each crime type, with 2003 as the reference year. Pre-2004 coefficients test the parallel trends assumption for this cohort. Design and sample identical to Table C.1 (13,120 individuals, 209,922 person-years). 5% confidence interval shown. Standard errors clustered at the grid level.

Table C.2: Impact of the mining boom on first-time and repeat offenders for 30-39 years old, 2000-2015

	(1) First-time convicted Property crime	(2) Re-offense Property crime	(3) First-time convicted Substance crime	(4) Re-offense Substance crime
Post*Treated	0.0002	-0.0000	-0.0006	-0.0002
	(0.0011)	(0.0007)	(0.0006)	(0.0004)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	209922	209922	209922	209922
N	13120	13120	13120	13120
Mean dep. var (2000-03)	0.0045	0.0010	0.0014	0.0005
Effect relative to the mean (%)	4.15	-4.84	-40.26	-49.88
R-squared	0.1874	0.3943	0.2161	0.4678
Within R-squared	0.0000	0.0000	0.0000	0.0000

Notes: The table replicates Table 2 for males aged 30–39. Outcome definitions (first-time convicted, re-offense) and design are identical to Table 2, with the sample replaced by 30–39-year-old male residents (13,120 individuals, 209,922 person-years). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3: Impact of the mining boom on criminal behavior by incarceration status for 30-39 years old, 2000-2015

	(1) Convicted + no prison Property crime	(2) Convicted + in prison Property crime	(3) Post-prison reoffense Property crime	(4) Convicted + no prison Substance crime	(5) Convicted + in prison Substance crime	(6) Post-prison reoffense Substance crime
Post*Treated	-0.0000	-0.0000	0.0002	-0.0007	0.0001	-0.0001
	(0.0011)	(0.0004)	(0.0006)	(0.0005)	(0.0003)	(0.0004)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Nxt	209922	209922	209922	209922	209922	209922
N	13120	13120	13120	13120	13120	13120
Mean dep. var (2000-03)	0.0041	0.0008	0.0007	0.0011	0.0004	0.0004
Effect relative to the mean (%)	-1.13	-3.56	37.07	-69.08	13.07	-28.58
R-squared	0.2311	0.2264	0.4059	0.3273	0.2387	0.4908
Within R-squared	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Notes: The table replicates Table 3 for males aged 30–39. Outcome definitions (convicted + no prison, convicted + in prison, post-prison reoffense) and design are identical to Table 3, with the sample replaced by 30–39-year-old male residents (13,120 individuals, 209,922 person-years). Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C.4: Impact of the mining boom on criminal behavior by distance to the mines for 30-39 years old, 2000-2015

	(1) Property crime	(2) Violent crime	(3) Traffic crime	(4) Substance crime
Post*≤ 20 km	-0.0002	0.0004	0.0012	-0.0007
	(0.0013)	(0.0011)	(0.0016)	(0.0007)
Post* 20 - 40 km	0.0018	-0.0029	-0.0074	-0.0033
	(0.0053)	(0.0027)	(0.0055)	(0.0038)
Post*40 - 60 km	0.0007	-0.0014	0.0032	-0.0005
	(0.0011)	(0.0023)	(0.0060)	(0.0004)
Post*60 - 80 km	-0.0003	-0.0017	-0.0034	-0.0003
	(0.0029)	(0.0026)	(0.0046)	(0.0004)
Individual FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Grid FE	Yes	Yes	Yes	Yes
Nxt	209922	209922	209922	209922
N	13120	13120	13120	13120
Mean dep. var (2000-03)	0.0055	0.0036	0.0067	0.0019
R-squared	0.3088	0.2396	0.3071	0.4397
Within R-squared	0.0000	0.0000	0.0000	0.0000

Notes: The table replicates Table A.9 for males aged 30–39. Treatment is defined by 20-kilometer distance rings from the nearest mine, with residents beyond 80 km as the reference. Design identical to Table A.9 with the sample replaced by 30–39-year-old male residents (13,120 individuals, 209,922 person-years). Crime definitions as in Table 1. Standard errors (in parentheses) clustered at the grid level. Data from administrative registers, Statistics Sweden (LISA). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

D Appendix: The causal forest approach for heterogeneous treatment effects

Using the causal forest method, I estimate the Conditional Average Treatment Effects (CATE) of the form:

$$CATE \equiv \tau(x) = E[Y_{1i} - Y_{0i}|X_i = x] \tag{3}$$

where Y_{1i} and Y_{0i} are the potential outcomes of interest for the ith individual when treated and untreated, respectively, and X is a vector of observable characteristics. The causal forest approach is a form of supervised machine learning techniques that is used for predicting heterogeneity in causal treatment effects (Athey and Imbens, 2016; Wager and Athey, 2018).[39] I follow the generalized random forest implementation developed by Athey et al. (2019). By using these methods, I rely on data-driven sample splits, thus limiting the researcher's discretion when selecting the relevant dimensions of heterogeneity. Given that I have a difference-in-differences setting (e.g., Davis and Heller, 2017; Britto et al., 2022), which is different than most applications based on RCTs, I run the causal forest over first differences, comparing pre- and post-boom averages. By doing this, the unconfoundedness assumption, explained in Wager and Athey (2018), holds because the treatment indicator is orthogonal to the covariates.

The method estimates conditional average treatment effects (CATEs), which are average treatment effects (ATEs) conditional on a set of variables for which the treatment effects may vary. I focus on two different estimates: individual average treatment effects (IATEs) and group average treatment effects (GATEs). IATEs are treatment effects conditional on observation-level characteristics, and there is one IATE for each observation in the sample. GATEs are treatment effects conditional on prespecified groups, and there is a treatment effect for each group. The approach fits an outcome model and a treatment-assignment model. I fit these models using cross-fitting via random forest. The CATEs are estimated using a partialing-out (PO) estimator via random forest. The algorithm randomly partitions the data across a large number of trees to flexibly capture heterogeneity in treatment effects without imposing a parametric structure. By default, the sample is randomly split into two parts ("honest" estimation): one half is used to determine the tree structure (e.g., how the data are partitioned into leaves), and the other half is used to estimate treatment effects within those leaves. This approach prevents overfitting and ensures unbiased estimation of treatment effects. The final CATE prediction for each observation is obtained by averaging over all trees in the forest. The default settings use 2000 trees, with subsampling and minimum leaf sizes chosen automatically by the algorithm to balance bias and variance. In addition, inference

[39]See Athey and Imbens (2019) for a review and discussion on recent machine learning (ML) literature for economics and econometrics.

and confidence intervals are computed using the bootstrap of little bags proposed in Athey et al. (2019).

In my specific case, the main outcomes are the probability of criminal conviction for property or substance-related crimes. The algorithm starts by building trees defined by data-driven sample splits characterizing leafs, which are followed by a prediction of the causal effect over the characteristics X. I believe that the treatment effect of the mining boom could vary based on schooling, earnings, employment status, and economic sectors, which I denote as x. Treatment(1) represents the potential outcomes of being treated, and Treatment(0) represents the potential outcomes of not being treated. I estimate the effects of the mining boom on criminal behavior conditional on the variables x:

$$IATE \equiv \tau(x) = E\{treatment(1) - treatment(0)|x\} \tag{4}$$

As x refers to individual characteristics, this version of the CATE is also known as IATEs. In this approach, I do not assume any functional form of $\tau(x)$, therefore, the data tells us what this function looks like.

If I want to know how the ATEs vary across population groups, I estimate the GATEs. Specifically, if G is a group variable (e.g., schooling levels) and g is a specific level of the group variable (e.g., primary education), I estimate the ATE conditional on belonging to group g, that is:

$$GATE \equiv \tau(g) = E\{treatment(1) - treatment(0)|G = g\} \tag{5}$$

where the function $\tau(g)$ is referred to as the GATE function.

E Appendix: Literature comparisons, resource shocks and crime

I compare the baseline estimated effect of the mining boom on criminal behavior with the effects of other resource shocks evaluated in the literature. To benchmark the findings, I calculate the effect sizes for related work against the control complier mean, the complier mean, the control mean, or the mean value of the criminal behavior measure, in that order of priority based on availability. I apply the same transformations to the confidence intervals. When the outcome is in log points, I interpret the effect as $100 \times (e^{\beta} - 1)$. Below, I detail this calculation for each paper included in the literature comparison plots in Online Appendix Figure B.1.

1. Andrews and Deza (2018) studies how a change in oil reserves in Texas impacts the crime in counties that have reserves. The authors exploit plausibly exogenous changes in the value of reserves and estimate reduced form models to capture the relationship between changes in the value of oil reserves and criminal activity in a given Texas county. As the independent variable of interest is an interaction between the oil price in the previous year and the amount of time-invariant reserves in million barrels of oil in any given county, I use the 26% increase in value of reserves reported in the paper to convert the results into comparable elasticities to the other papers using a DID. As outcomes, the authors have several types of crime. For simplicity, I focus on murder and robbery as proxies for violent and property crimes. The authors find that a 1% increase in the value of oil reserves increases murder by 0.16% and robbery by 0.55%. Using the 26% increase in the value of oil reserves to convert the results, there is a 4.1% (95% CI: 0.5% to 7.8%) increase in murder ([Table 2, Column 1]). Moreover, there is a 14.3% (95% CI: 2.8% to 25.9%) increase in robbery ([Table 2, Column 1]).

2. James and Smith (2017) studies how the energy boom of oil and shale gas in the United States affected regional crime rates throughout the country. The authors use a difference-in-differences design comparing counties for which the geographic center lies above one of the major play formations (treated) against controls, and exploiting the national temporal variation in shale energy production. They find positive effects on rates of various property and violent crimes in shale-rich counties. Focusing on all crimes, the authors find that the shock increased crime in treated counties by 0.080, significant at the 1% level ([Table 3, Column 6]. As the outcome is in log points, I interpret that there is an 8.3% (95% CI: 2.3% to 14.7%) increase in all crime.

3. Corvalan and Pazzona (2019) studies the short- and medium-run effects that an increase in copper price had on the local economy and on criminal activity in Chile. The authors compute the current value of the copper production in the year 2000 in each

municipality, in billions of Chilean pesos, and multiply it by the current price of copper in billions of Chilean pesos. Then, by comparing mining and non-mining municipalities, the authors find that, after a decade of high prices, mining municipalities did not exhibit lower crime rates compared to non-mining municipalities. As an outcome, the authors focus on property crimes and use the number of crime reports to the authorities per 100,000 inhabitants. As the independent variable of interest is an interaction between the copper production in the year 2000 and the price, I use the 400% increase in the international price of copper reported in the paper to convert the results into comparable elasticities to the other papers using a DID. The authors find that a 1 billion CLP increase in the value of copper production reduces property crime by 0.98 per 100,000. Using the 400% increase in the price to convert the results, there is a 1.2% (95% CI: -3.5% to 1.1%) reduction in property crime, which is not statistically significant ([Table 3, Column 6]).

4. Axbard et al. (2021) studies the impact of natural resource wealth on criminal activity in South Africa. The authors exploit price fluctuations in 15 internationally traded minerals as exogenous variation and compare mining police precincts against controls. The outcome of interest is the inverse hyperbolic sine transformed total number of crimes. As the independent variable of interest is the inverse hyperbolic sine transformation of the mineral value, I use the 154% increase in mining value reported in the paper to convert the results into comparable elasticities to the other papers using a DID. The authors find that increased mineral wealth leads to less crime. Specifically, the authors find that a 10% increase in the value of mineral production reduces the total number of crimes by about 0.7% (significant at the 5%-level). Using the 154% increase in mining value to convert the results, there is a 11.1% (95% CI: -20.4% to -1.7%) crime reduction ([Table 1, Column 1]).

5. Komarek (2018) studies the effect of resource extraction on local crime using the fracking boom as a natural experiment in the Marcellus region in the United States. The author uses a difference-in-differences model, exploiting variation in both the timing of fracking activity in a county and the moratorium on fracking natural gas in the State of New York. That is, counties in Pennsylvania can receive the treatment of fracking activity, while similar counties in New York can only serve as controls due to the policy. He finds that areas experiencing a natural gas extraction boom suffer an increase in overall violent crimes, while property crimes remain similar to non-boom areas. Specifically, the author finds that the shock increased violent crime in treated counties by 0.288, significant at the 1% level ([Table 2, Column 2]). For property crimes, the author finds that the shock increased property crime in treated counties by 0.050, which is not statistically significant ([Table 2, Column 4]). As the outcome is transformed using the inverse hyperbolic sine transformation of the number of crimes

per 100,000 residents, I interpret that there is a 33.4% (95% CI: 10.8% to 60.5%) increase in violent crimes and a 5.1% (95% CI: -3.2% to 14.2%) increase in property crimes.

6. Street (2025) studies the effect of the fracking boom in North Dakota, both at the individual and aggregate levels, on criminal behavior. The author uses a generalized difference-in-differences framework, comparing the criminal behavior of resident households in counties within the shale play to residents in counties outside the shale play, before and after the fracking boom. The author considers two periods: leasing (2004–2008) and production (2008–2017). I compare my effects with the effects of the production period. At the aggregate level, the outcome is aggregate cases and charges filed per household population for each county-year, and the author finds large increases in charges and cases filed during the production period. Specifically, there is a 0.0371 percentage point increase in cases per household during the production period, translating to a 44.7% (95% CI: 13.3% to 75.9%) increase, using the baseline mean of 0.083, significant at the 5% level ([Table 3, Column 1]). At the individual level, the outcome is a binary indicator for whether a case was filed for the household each year, and evidence shows a modest decrease in crime for treated individuals. Specifically, there is a 0.35 percentage point decrease during the production period in the probability of having a case filed for treated individuals, translating to a 17.5% (95% CI: -34.2% to -0.8%) decrease, using the baseline mean of 0.02, significant at the 5% level ([Table 2, Column 2]).

F Appendix: Newspaper content analysis

F.1 Data and search procedure

I use the Retriever Mediearkiv, a comprehensive Swedish newspaper archive that covers the large majority of Swedish daily and regional newspapers and is the standard archive in Swedish media research (accessible through the BIBSAM consortium university license). I search for all articles containing the string `LKAB*`, where the wildcard captures inflected forms such as *LKABs* (genitive) and compound words such as *LKAB-anställda* (LKAB employees). The search covers January 2000 to December 2015, yielding 16 years of coverage that span four pre-boom years (2000–2003) and twelve post-boom years (2004–2015).

Because Retriever limits each export to 500 articles, articles were downloaded in monthly or bi-monthly batches for high-volume years and combined. After restricting to articles with a valid publication date and non-empty body text, the final sample contains approximately 16,569 articles. Figure 1, panel (b), shows the rapid increase in annual article counts following the mining boom, reflecting the company's growing newsworthiness.

F.2 Outcome variable and econometric model

For each calendar month t, I construct the share of articles mentioning at least one term from a given dictionary:

$$S_t = \frac{\text{articles mentioning keyword(s) in month } t}{\text{total LKAB articles in month } t}.$$

Using a share rather than a count controls for the large post-boom increase in total LKAB coverage, isolating whether the specific topic became a larger fraction of the conversation. Matching is case-insensitive substring matching, which is important in Swedish because of compound-word formation (e.g., *arbetsmiljöproblem* would be missed by exact-word matching).

I estimate a monthly interrupted time series (ITS) model of the form:

$$S_t = \alpha + \beta_1 \text{Post}_t + \beta_2 t + \beta_3 (t \times \text{Post}_t) + \varepsilon_t, \tag{6}$$

where S_t is the share of articles classified in a specific topic in month t, the variable $\text{Post}_t = 1$ for $t \geq$ January 2004 indicates the post-boom period, and t is a running variable measured in months, equal to zero at January 2004, negative before, and positive after. I include $t \times \text{Post}_t$ to allow for potential change in the trend following January 2004, which captures the slope specific to the post-boom period. The coefficients are interpreted as follows: β_1 represents the immediate level shift in the share in January 2004, which is the primary coefficient of interest. β_2 is the pre-boom monthly time trend, β_3 is the change in trend after the boom, and the post-boom slope is $\beta_2 + \beta_3$. α is the baseline share in January 2004. The error term is captured by ε_t. Standard errors follow Newey and West (1987) with 12 lags, which is the standard choice for monthly data (Ghazarian, 2025), and correct for both serial correlation and heteroskedasticity.

Identification. The ITS design relies on the assumption that, absent the mining boom, the share S_t would have continued along its pre-2004 trend. This "parallel trend in time" assumption cannot be tested directly. The design has no control group; therefore, concurrent national trends in the discourse or changes in newspaper coverage of the specific topic more broadly are not differenced out. Results should therefore be interpreted as descriptive and suggestive rather than causal. Additional caveats include the short pre-period (48 monthly observations), keyword false positives arising from compound words, and potential changes in the mix of newspapers covering LKAB over time.

F.3 Topic decomposition of LKAB coverage

General topic coverage of LKAB. To characterize what the mining boom looked like in public discourse, I perform the keyword analysis on seven broad topic categories: production and economic output, employment and hiring, accidents and safety, environment and community, labor relations, stress and mental health, and crime. For each article, I flag membership in each topic category using case-insensitive substring matching; an article may be assigned to multiple topics. Table F.1 lists the keyword dictionary.

Table F.1: Topic keyword dictionary (Swedish)

Topic	**Swedish keywords**	**English meaning**
Production & economy	produktion, järnmalm, export, investering, vinst, pris, marknad, efterfrågan, expansion, utbyggnad, gruvdrift	production, iron ore, export, investment, profit, price, market, demand, expansion, mine operations
Employment & hiring	anställa, rekrytera, arbetstillfälle, lediga jobb, nyanställd, nyanställning, arbetskraft, bemanning	hire, recruit, job opportunity, job opening, new hire, workforce, staffing
Accidents & safety	olycka, olyckor, skadad, dödsolycka, arbetsolycka, gruvras, bergras, jordskalv, marksättning, driftstopp, driftstörning, produktionsstopp	accident, injured, fatal accident, mine collapse, rockslide, earthquake, ground subsidence, shutdown
Environment & community	miljö, utsläpp, damm, buller, samhällsomvandling, stadsomvandling, marksubsidenser, renar, samer, sameby	environment, emissions, dust, noise, urban transformation, city relocation, reindeer, Sami
Labor relations	strejk, konflikt, fack, löneförhandling, varsel, uppsägning, nedskärning	strike, conflict, union, wage negotiation, redundancy notice, dismissal, cutback
Stress & health	stress, utbränd, utmattning, sjukskrivning, psykisk, arbetsmiljö, depression, ångest, ohälsa, välmående	stress, burnout, exhaustion, sick leave, psychological, work environment, depression, anxiety, wellbeing
Crime	brott, kriminalitet, stöld, narkotika, polisen, dömd, fängelse, rättegång, åtalad, häktad	crime, theft, narcotics, police, convicted, prison, trial, prosecuted, detained

Note: The table lists the Swedish-language terms used to classify LKAB-related newspaper articles by topic. An article is assigned to a topic if it contains at least one term from that topic's dictionary. Matching is case-insensitive substring matching to capture Swedish compound words. Terms excluded to avoid false positives: *kiruna*, *gällivare*, *malmberget* (place names appearing in virtually all LKAB articles); *gruvan* (generic "the mine"); *personal* (generic "staff/personnel"); *ras* (homonym: also means "race/breed"); *seismisk* (ubiquitous for mine subsidence); *evakuering* (Kiruna city relocation, present in a large share of all articles); *stopp* (too broad). An article may be assigned to multiple topics.

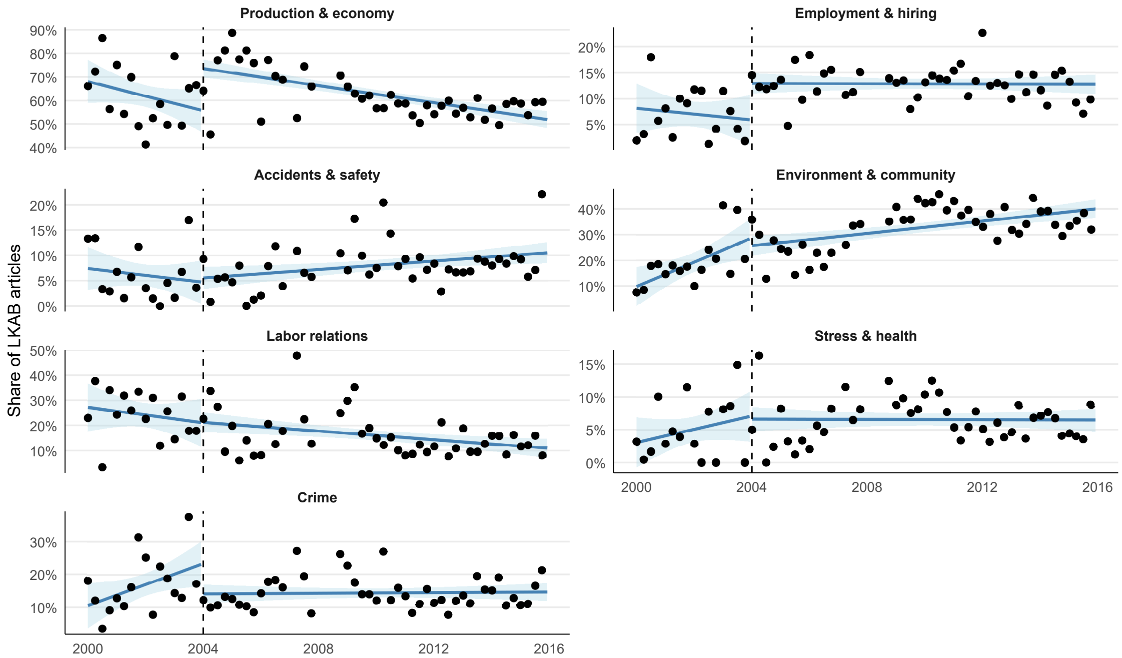
Table F.2 reports the ITS estimates for the different topics alongside pre- and post-boom mean shares. Figure F.1 plots the pre- and post-boom ITS fitted trends for all seven topics. The only topics showing statistically significant level shifts at the boom onset are employment and hiring and production and economy, consistent with boom-driven news centered on expansion and hiring. By contrast, stress and health, accidents and safety, and labor relations show no significant level shift, while crime coverage falls significantly, in line with the reduction in property crime convictions.

Table F.2: Topic decomposition of LKAB newspaper articles, 2000–2015

	Pre-boom (2000–2003)	Post-boom (2004–2015)	Post-period indicator ($\hat{\beta}_1$)	p-value
Production & economy	61.9%	62.4%	0.180**	0.022
			(0.078)	
Employment & hiring	7.0%	12.8%	0.071***	0.000
			(0.013)	
Accidents & safety	6.1%	8.1%	0.008	0.750
			(0.026)	
Environment & community	19.2%	33.1%	-0.031	0.497
			(0.045)	
Labor relations	24.2%	15.9%	0.002	0.967
			(0.039)	
Stress & health	5.0%	6.5%	-0.005	0.845
			(0.024)	
Crime	16.9%	14.4%	-0.094***	0.000
			(0.025)	

Notes: The table reports estimates from the interrupted time series (ITS) model for the different topics. The outcome is the monthly share of LKAB*-related newspaper articles mentioning each topic. The ITS model estimates three parameters: an immediate level shift at January 2004, a pre-boom monthly linear trend, and a change in slope after the boom. The table shows the immediate level shift at January 2004. The sample covers 175 monthly observations (2000–2015). Newey–West standard errors (in parentheses), lag = 12. Data from Retriever Mediearkivet. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure F.1: Interrupted time series, topic decomposition of LKAB newspaper articles, 2000–2015



Notes: Each panel plots a quarterly binscatter of the monthly share of LKAB*-related articles mentioning keywords from the indicated topic category over 2000–2015. The fitted lines represent the estimated pre- and post-boom linear trends from the interrupted time series regression. The vertical dashed line marks January 2004, which corresponds to the beginning of the mining boom. Articles are identified using the newspaper archive *Retriever Mediearkivet*. Following the literature, we search for articles containing the case-insensitive string "LKAB*", where the asterisk is used as a wildcard. LKAB is the main iron ore producer in Sweden and operates the large-scale mines in Kiruna and Gällivare. LKAB is Sweden's state-owned iron ore mining company. Newey-West standard errors (lag = 12).

Stress coverage of LKAB. To test the occupational stress channel, I perform the keyword analysis in more detail for the stress and mental health topic. Table F.3 shows the stress and mental health dictionary, listing the terms used to classify articles as mentioning this topic and the rationale for why they are used.

Table F.4 reports the ITS estimates for the stress series under three specifications: a baseline linear trend (column (1)), a robustness check that adds a quadratic time term (column (2)), and a specification that weights each monthly observation by the total number of articles in that month (column (3)). In the baseline specification, the estimated level shift at January 2004 is $\hat{\beta}_1 = -0.007$, indicating no statistically significant change in the share of LKAB articles mentioning stress or mental health terms at the onset of the boom. The pre-boom trend ($\hat{\beta}_2 = 0.001$ and the post-boom slope change ($\hat{\beta}_3 = -0.001$) are both positive and negative, respectively, but neither is statistically significant, providing no evidence of a systematic trend in stress-related coverage before or after the boom. Column (2) adds a quadratic time term; the level shift remains negative and statistically insignificant. I follow Ghazarian (2025) in using this quadratic robustness check as a standard test of the functional form assumption on the pre-period trend specification. Finally, Column (3) weights

Table F.3: Stress and mental health keyword dictionary (Swedish)

Swedish term	English meaning	Rationale
stress / stressad	stress / stressed	Core term
utbränd / utbrändhet	burned out / burnout	Clinical work-related exhaustion
utmattning	exhaustion / fatigue	Physical and mental exhaustion
sjukskrivning	sick leave (certified)	Objective indicator of worker health
sjukfrånvaro	sick absence	Absence due to illness
psykisk	mental / psychological	Modifier for health/illness
psykosocial	psychosocial	Occupational health technical term
ohälsa	ill-health / poor health	General health problems
arbetsmiljö	work environment	Central to labor law discussions
arbetsbelastning	load / workload	Work overload (compound only)
depression	depression	Clinical mental health condition
ångest	anxiety	Co-occurs with stress and substance use
välmående / välbefinnande	wellbeing	Worker wellbeing

Note: The table lists the Swedish-language terms used to classify LKAB-related newspaper articles as mentioning occupational stress or mental health, with English translations and rationale. Matching is case-insensitive substring matching to capture Swedish compound words. Two terms are excluded to avoid false positives: "press" (Swedish for "media/press", appearing in virtually all news articles) and standalone "belastning" ("load/strain", common in mining engineering contexts); the compound "arbetsbelastning" ("work overload") is retained as it is specific to occupational health. The monthly share variable is the fraction of LKAB articles in a given month containing at least one of these terms.

observations by the total number of articles per month (Ghazarian, 2025); the result is again insignificant, suggesting that the baseline result is not driven by months with sparse coverage. Taken together, the three specifications provide no evidence of a structural break in stress-related media coverage at the onset of the mining boom. Figure 6 (in the main text) presents the binscatter of quarterly shares alongside the ITS fitted trends of Column (1).

Table F.4: Interrupted Time Series: Share of LKAB articles mentioning stress/mental health terms, 2000-2015

	(1) Baseline	(2) Quadratic trend	(3) Weighted
Post-period indicator (β_1)	-0.007	-0.023	0.031
	(0.024)	(0.029)	(0.025)
Time in months (β_2)	0.001	0.001	0.001
	(0.001)	(0.001)	(0.001)
Time^2		-0.000	
		(0.000)	
Post-period slope (β_3)	-0.001	0.000	-0.001*
	(0.001)	(0.001)	(0.001)
Constant	0.075***	0.073***	0.066***
	(0.019)	(0.019)	(0.014)
Observations	175	175	175
R-squared	0.024	0.043	0.051

Notes: The table reports estimates from the interrupted time series (ITS) model underlying Figure 6. The outcome is the monthly share of LKAB*-related newspaper articles mentioning stress or mental health terms (keyword dictionary in Table F.3). The ITS model estimates three parameters: an immediate level shift at January 2004, a pre-boom monthly linear trend, and a change in slope after the boom. Column (1) is the baseline; Column (2) adds a quadratic time trend as a functional form check; Column (3) weights observations by article count per month. The sample covers 175 monthly observations (2000–2015). None of the coefficients is statistically significant in any specification. Newey–West standard errors (in parentheses), lag = 12. Data from Retriever Mediearkivet. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

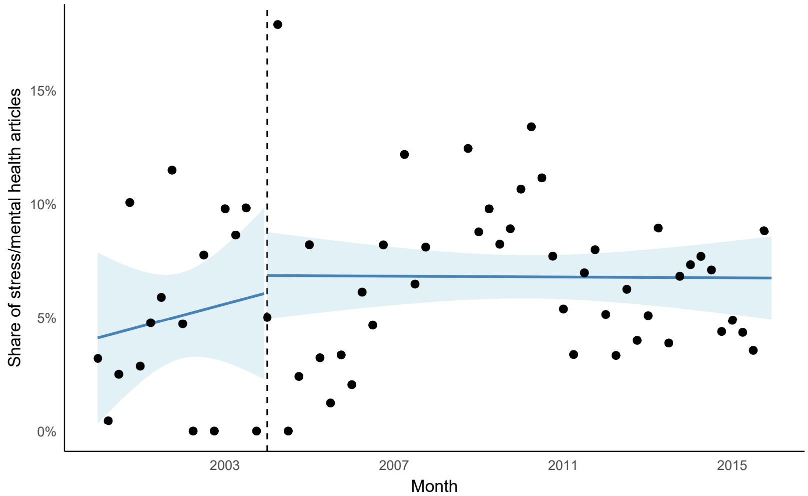
As a robustness check, I implement a minimum monthly article threshold. Pre-boom months contain substantially fewer total LKAB articles than post-boom months. In months with very few articles, a single article mentioning a stress term can shift the monthly share substantially, inflating pre-boom variance. To assess whether this drives any results, I drop months with fewer than five total articles and re-estimate the ITS. Table F.5 reports the results and Figure F.2 plots the series. The estimated level shift at January 2004 is $\hat{\beta}_1 = 0.008$, and neither the pre-boom trend ($\hat{\beta}_2 = 0.000$) nor the post-boom slope change ($\hat{\beta}_3 = 0.000$) is statistically significant. The standard errors on β_1 shrink from 0.024 (full sample) to 0.021 (restricted sample), confirming that dropping sparse months increases precision without affecting the null result.

Table F.5: Interrupted Time Series: Share of LKAB articles mentioning stress/mental health terms (N≥5 robustness), 2000-2015

	(1) Baseline	(2) Quadratic trend	(3) Weighted
Post-period indicator (β_1)	0.008	-0.009	0.035
	(0.021)	(0.026)	(0.024)
Time in months (β_2)	0.000	0.000	0.001
	(0.000)	(0.000)	(0.001)
Time^2		-0.000	
		(0.000)	
Post-period slope (β_3)	-0.000	0.001	-0.001
	(0.000)	(0.001)	(0.001)
Constant	0.061***	0.058***	0.063***
	(0.014)	(0.013)	(0.014)
Observations	173	173	173
R-squared	0.021	0.041	0.052

Notes: The table replicates Table F.4 after excluding months with fewer than five total LKAB articles, addressing concerns that sparse pre-boom months (where a single stress-related article can inflate the share) may drive results. The restricted sample covers 173 monthly observations. Column specifications are identical to Table F.4; all coefficients remain statistically insignificant. Newey–West standard errors (in parentheses), lag = 12. Data from Retriever Mediearkivet. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure F.2: Interrupted time series of stress/mental health newspaper articles, months with $N \geq 5$ (robustness), 2000–2015



Notes: The figure plots a binscatter of the quarterly share of LKAB* articles mentioning stress or mental health terms from 2000 to 2015. Restricted to months with at least five total LKAB articles. Months with fewer than five articles are concentrated in the pre-boom period (2000–2003). The fitted lines represent the estimated pre- and post-boom linear trends from the interrupted time series regression. The vertical dashed line marks January 2004, which corresponds to the beginning of the mining boom. Articles are identified using the newspaper archive *Retriever Mediearkivet*. Following the literature, we search for articles containing the case-insensitive string "LKAB*", where the asterisk is used as a wildcard. LKAB is the main iron ore producer in Sweden and operates the large-scale mines in Kiruna and Gällivare. LKAB is Sweden's state-owned iron ore mining company. Newey-West standard errors (lag = 12). See Online Appendix Table F.5 for coefficient estimates and Online Appendix F for data and estimation details.

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